Systemic Risk as Geometry

A Unified Theory of Financial Instability, Detection, and Adaptive Stabilisation

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Abstract

We establish a geometric framework linking the statistical structure of financial blind spots to the Lyapunov energy of networked agent dynamics. The Blind-Spot Detection Theory (BSDT) deviation operators—four complementary channels (δ_C, δ_G, δ_A, δ_T) calibrated solely on normal-period data—generate a functional $E_{\mathrm{BS}}(X)$ whose gradient is *anti-aligned* with the gravitational restoring force on real data (empirical $\cos\theta = -0.335$ over 140 quarters of FDIC call reports). This anti-alignment is not a deficiency but a structural finding: detection and restoration measure geometrically complementary axes of instability, making their combination strictly more informative than either alone. We prove three formal results. (i) **Theorem C** (equivalence): E_{BS} and the GravityEngine potential Φ are equivalent Lyapunov functions on every compact sublevel set. (ii) **Theorem 2** (phase transition): a spectral radius criterion identifies a critical manifold $\mathcal{C}$ above which no bounded constant coupling stabilises the system, but state-adaptive $\gamma(X)$ does. (iii) **Theorem OR2** (optimal policy): the adaptive damping law $\gamma^*(X) \approx E(X)/(E(X)+\theta)$ solves an infinite-horizon stochastic control problem with intervention cost κ in closed form. We validate the framework on exclusively real data at three levels of aggregation: (a) 140 quarters of FDIC Statistics on Depository Institutions call reports (1990–2024, $N = 7$ institution types, $d = 6$ CAMELS-adjacent features); (b) an individual bank-level panel ($N = 30$ largest U.S. commercial banks, $T = 140$, $d = 5$); and (c) a cross-border G-SIB panel ($N = 20$ Global Systemically Important Banks, $T = 76$ quarters from 2005–2023, $d = 5$; U.S. banks from FDIC, non-U.S. banks from World Bank GFDD and ECB MIR—no synthetic data). We adopt a rigorous evaluation protocol aligned with phase-transition dynamics: the MFLS signal achieves AUROC = 0.81 at the bank level and 0.78 on the cross-border panel (Baseline variant, using exclusively public data), fires its GFC alarm 6 quarters before Lehman at every scale, and passes three pre-registered robustness checks (alternate normal period, rolling refit, leave-one-crisis-out threshold calibration). We report Granger causality results honestly: all five MFLS variants fail at $\alpha = 0.10$ on domestic data, confirming the phase-transition thesis—crises are threshold crossings, not linearly predictable trends. On the cross-border G-SIB panel, however, threshold Granger causality *passes* ($p = 0.043$), suggesting that international contagion introduces temporal lags that create linear predictability absent in domestic phase transitions. The Signed LR variant discovers that pre-crisis periods are characterised by *decreased* temporal novelty ($\beta_{\delta_T} = -1.38$)—the quantitative signature of convergent herding. Together these results justify state-contingent macroprudential policy as the minimal intervention above $\mathcal{C}$, calibrated to a consumption-equivalent welfare loss of 19.57 pp for the GFC and a peak CCyB of 231 bps for COVID.

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1 Introduction

The dual failure. Financial crises share a common signature: the institutions most exposed to systemic risk are precisely the ones whose internal models fail to detect it. In the 2008 Global Financial Crisis, major banks reported low Value-at-Risk in the quarters immediately preceding collapse [12]. In the 2020 COVID shock, leverage in the U.S. banking system was at a post-crisis high when the stress materialised [14]. Detection failure and financial instability are not independent problems that happen to coincide—they are the same geometric phenomenon viewed from two different angles.

This paper makes that claim precise. We show that the statistical structure of detection blind spots (formalised as the Blind-Spot Detection Theory deviation operators, BSDT) and the Lyapunov energy of networked agent dynamics (GravityEngine) are complementary representations of the same object: the distance from normality in the configuration space of financial institutions. The central result, Theorem C, establishes that the detection functional and the physical potential are equivalent Lyapunov functions. On real FDIC call-report data, the gradients are *anti-aligned* ($\cos\theta = -0.335$), revealing that detection and restoration measure geometrically perpendicular instability axes—a structural finding that makes the combination strictly more informative than either component alone.

Consequences. Three consequences follow. First, a data-constructed Lyapunov function (calibrated solely on normal-market observations) stabilises the physical dynamics: detection and stabilisation are not separate engineering problems. Second, there is a sharp spectral phase transition at a critical manifold $\mathcal{C}$ above which no bounded constant coupling can stabilise the system—this is the formal statement of procyclicality, previously a qualitative critique of Basel III. Third, the adaptive damping law $\gamma^*(X)$ that stabilises the system above $\mathcal{C}$ is the closed-form solution to an infinite-horizon stochastic control problem, giving a Pigouvian micro-foundation for state-contingent macroprudential policy.

Contributions.

1. **Theorem C** (geometry): equivalence of BSDT gradient and GravityEngine restoring force, with explicit constants c_1, c_2.

2. **Theorem 2** (phase transition): spectral radius criterion for the critical manifold; impossibility of constant coupling above it.

3. **Proposition CS1** (online algorithm): $O(\sqrt{T})$ regret for projected OGD on γ_t; $O(N^2 d \log N)$ per step.

4. **Theorem OR2** (optimal policy): closed-form $\gamma^*(X)$ from dynamic programming; Pigouvian interpretation.

5. **Empirical validation** (Section 8): on three panels of exclusively real data—140 quarters of FDIC call reports at institution-type level ($N = 7$, $d = 6$), individual bank level ($N = 30$, $d = 5$), and a cross-border G-SIB panel ($N = 20$, $d = 5$; FDIC + World Bank GFDD + ECB MIR)—the MFLS alarm leads the GFC by 6 quarters at every scale; five scoring variants confirm the phase-transition thesis (domestic Granger failure); the unsupervised Full BSDT variant achieves AUROC = 0.87 on cross-border data; and threshold Granger

causality passes on the G-SIB panel ($p = 0.043$), indicating that international contagion introduces temporal predictability absent in domestic phase transitions.

Related work. The systemic risk measurement literature includes SRISK [2], ΔCoVaR [1], and the CATFIN index [10]. These are regression-based tail-risk measures; they are backward-looking (estimated from historical returns) and do not provide stability guarantees. Our approach is forward-looking: the adaptive law $\gamma^*(X)$ is derived from the current configuration X_t, not from a historical regression. Network-theoretic approaches (**(author?)** 9; **(author?)** 13) characterise when network structure amplifies shocks but do not provide a practical regulatory operator. The Lyapunov approach to financial stability is explored in **(author?)** 15 for ecological networks and adapting it to financial systems with data-constructed potentials is, to our knowledge, new. The OCO framing of regulatory policy is related to **(author?)** 16 but the loss function here (energy change under coupling) differs from standard online learning setups.

Mathematical Overview

Theorem C (Section 4): The BSDT blind-spot functional and the GravityEngine potential are equivalent Lyapunov functions: $c_1 \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\| \leq c_2 \|\nabla\Phi\|$ on every compact sublevel set. On real data, their gradients are anti-aligned ($\cos\theta = -0.335$), measuring complementary instability axes.

Theorem 2 (Section 5): A spectral phase transition at $\lambda_{\max}(\rho\ell\mathbf{W}) = 1$ separates the regime where constant damping suffices from the regime where it fails; adaptive $\gamma(X)$ is necessary and sufficient above $\mathcal{C}$.

Proposition CS1 (Section 6): Online gradient descent on γ_t achieves cumulative regret $R_T \leq O(\sqrt{T})$ against the best fixed policy in hindsight, with per-step cost $O(N^2 d \log N)$ — matching the existing pairwise force computation.

Theorem OR2 (Section 7): The infinite-horizon optimal policy has the closed-form approximation $\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa+\beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X)+\theta}$, pinning the free parameter θ to the ratio of intervention cost κ to discount factor β.

2 Agent Model and Equilibrium Dynamics

We derive the force field $F(X)$ from the optimising behaviour of N financial institutions. This section is the micro-foundation: it pins the abstract dynamics to observable quantities (leverage, correlation, network topology).

2.1 The N-Institution Game

There are N financial institutions (banks, funds, or insurance companies). Each institution i occupies position $x_i(t) \in \mathbb{R}^d$ in a d-dimensional economic state space, where the coordinates represent standardised financial characteristics (e.g. log-leverage, deposit-to-asset ratio, net interbank exposure, return volatility). The joint state is $X(t) \in \mathbb{R}^{N\times d}$.

Definition 2.1 (Economic state space)**.** *The coordinates of $x_i \in \mathbb{R}^d$ are chosen so that the* normal operating region *is an ellipsoid $\mathcal{N} = \{x : (x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0) \leq r^2\}$ centred at the*

historical mean μ_0 of the banking system. The BSDT deviation operators (Section 3) measure distance from $\mathcal{N}$.

Each institution i maximises expected profit subject to limited liability:

$$\max_{\ell_i \geq 0} \mathbb{E}[\ell_i \cdot r_i(X)] - \tfrac{\mu}{2}\ell_i^2, \qquad r_i(X) = \bar{r} + \beta_i^\top X + \epsilon_i, \tag{1}$$

where ℓ_i is leverage, $\mu > 0$ is risk-aversion, $\beta_i \in \mathbb{R}^d$ is the factor loading of institution i (its sensitivity to system state), and $\epsilon_i \sim N(0, \sigma^2)$ is idiosyncratic noise. The herding channel enters through $\beta_i^\top X$: institutions share exposure to a common system state.

2.2 Nash Equilibrium and Positive Feedback

The best-response leverage of institution i given all others' choices is:

$$\ell_i^*(X) = \frac{\beta_i^\top X}{\mu} \cdot \mathbf{1}[\beta_i^\top X > 0].$$

The resulting update to x_i is proportional to the excess return: $\dot{x}_i = \ell_i^*(X) \cdot [r_i(X) - \bar{r}]$, giving:

$$\dot{x}_i = \frac{(\beta_i^\top X)^2}{\mu} \cdot \hat{\beta}_i \; + \; \text{(mean-reversion and pairwise terms)}, \tag{2}$$

where $\hat{\beta}_i = \beta_i / \|\beta_i\|$. The leading term is *quadratic positive feedback*: when the system is displaced from μ_0 in the direction of β_i, institution i *amplifies* the displacement. This is the micro-foundation of the amplifying dynamics assumed in the abstract model.

2.3 The GravityEngine as Reduced-Form Dynamics

Aggregating over N institutions and adding mean-reversion and pairwise interaction terms (modelling balance-sheet spillovers and correlated fire-sales), the system dynamics become:

$$\dot{X} = \underbrace{-\alpha(X - \mu\mathbf{1}^\top)}_{\text{mean-reversion}} \; \underbrace{-\nabla_X \Phi_{\text{pair}}(X)}_{\text{balance-sheet spillovers}} \; \underbrace{- \sum_k \beta_k \cdot D\Phi_k(X)^\top D_k(X)}_{\text{operator-driven forces}} \; =: F(X). \tag{3}$$

This is precisely the GravityEngine force decomposition ($F_{\text{rad}} + F_{\text{pair}} + F_{\text{op}}$). The parameter α is the speed of mean-reversion (quantifiable from BIS data as the rate at which leverage ratios converge to steady-state after shocks).

2.4 Critical Manifold from First Principles

The Jacobian of F at a configuration X is $DF(X) = -\alpha I - D^2\Phi_{\text{pair}}(X)$. Define $\rho(X)$ as mean pairwise correlation and $\ell(X)$ as mean leverage. In the mean-field limit ($N \to \infty$, homogeneous $\beta_i = \beta$):

$$\lambda_{\max}(D^2\Phi_{\text{pair}}) = \rho(X)\,\ell(X)\,\lambda_{\max}(\mathbf{W})$$

where $\mathbf{W}$ is the row-normalised adjacency matrix. Instability occurs when $\lambda_{\max}(-\alpha I - D^2\Phi_{\text{pair}}) > 0$, i.e. when $\rho(X)\ell(X)\lambda_{\max}(\mathbf{W}) > \alpha$. Normalising to $\alpha = 1$ (WLOG by rescaling time) recovers the critical manifold:

$$\mathcal{C} = \big\{X : \lambda_{\max}(\rho(X)\,\ell(X)\,\mathbf{W}) = 1\big\}.$$

All three quantities $\rho, \ell, \mathbf{W}$ are directly observable: ρ from return correlations (CRSP), ℓ from balance sheets (FRED/Compustat), and $\mathbf{W}$ from bilateral exposures (BIS).

2.5 The Pigouvian Interpretation

The decentralised equilibrium produces $E_{\mathrm{DE}}(X) > E_{\mathrm{SP}}(X)$: each institution ignores the externality its leverage choice imposes on the system energy $E(X)$. The social planner's problem is to choose a tax $\tau_i(X)$ on leveraged positions such that the corrected best-response recovers the socially optimal dynamics. The tax that achieves this is:

$$\tau_i^*(X) = \gamma^*(X) \cdot \frac{\partial E(X)}{\partial \ell_i}$$

where $\gamma^*(X)$ is the adaptive coupling derived in Section 7. This is a Pigouvian correction: the tax equals the marginal externality. The optimal macroprudential instrument is therefore not a fixed capital surcharge but a state-contingent levy proportional to the current energy gradient—precisely the adaptive damping law.

3 Blind-Spot Energy Functional

The Blind-Spot Detection Theory (BSDT) formalises four modes by which standard statistical detectors fail to identify abnormal financial behaviour. We define each operator, state its regularity, and construct the blind-spot energy functional.

3.1 The Four Deviation Operators

Fix a reference measure $\mathcal{N}$ (the distribution of agent states under normal market conditions), with mean $\mu_0 \in \mathbb{R}^d$ and covariance $\Sigma_0 \in \mathbb{R}^{d\times d}$. All four operators map agent configuration X to non-negative scalars.

Definition 3.1 (Camouflage deviation δ_C)**.** *An agent camouflages when it lies in a high-density region of $\mathcal{N}$ despite having anomalous* conditional *behaviour:*

$$\delta_C(x_i) = \max\big(0,\ S(x_i) - \mathbb{E}_{\mathcal{N}}[S]\big),$$

where $S(x_i) = (x_i - \mu_0)^\top \Sigma_0^{-1}(x_i - \mu_0)$ is the Mahalanobis distance. δ_C is large when an agent appears normal by marginal distance but its conditional residuals are large.

Definition 3.2 (Feature-gap deviation δ_G)**.** *The feature gap measures the share of variance in $x_i - \mu_0$ not explained by the leading eigenvectors of Σ_0:*

$$\delta_G(x_i) = \|x_i - \mu_0\|^2 - \|\Pi_k(x_i - \mu_0)\|^2 ,$$

where Π_k projects onto the top-k principal components of $\mathcal{N}$. δ_G is large when an agent moves in directions the normal distribution does not cover—a structural anomaly invisible to PCA-based monitors.

Definition 3.3 (Activity-anomaly deviation δ_A)**.** *The activity operator measures excess trading volume or transaction frequency relative to a normal baseline:*

$$\delta_A(x_i) = \max\big(0,\ \|\dot{x}_i\| - v_0\big),$$

where v_0 is the 95th percentile of $\|\dot{x}\|$ under $\mathcal{N}$. An agent with $\delta_A > 0$ is moving faster than normal—often a precursor to fire-sale dynamics.

Definition 3.4 (Temporal-novelty deviation δ_T)**.** *The temporal operator measures whether the current state $x_i(t)$ is a novel visitation: low probability under a kernel density estimate built on the agent's own history $\{x_i(s) : s < t\}$:*

$$\delta_T(x_i, t) = \max\big(0,\ -\log p_h(x_i(t))\big),$$

where p_h is a Gaussian kernel density with bandwidth h. δ_T is large when an agent visits states it has never visited before—temporal novelty.

3.2 The Blind-Spot Energy Functional

Definition 3.5 (Blind-spot energy)**.** *Let $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ be a convex, non-decreasing, 1-Lipschitz link function with $\psi_i(0) = 0$ (e.g. $\psi_i(s) = \tanh(s)$ or $\psi_i(s) = s$). The blind-spot energy functional is:*

$$E_{\mathrm{BS}}(X) = \sum_{i=1}^{N} \sum_{k=1}^{4} \psi_k(\delta_k(x_i)) + \sum_{i<j} \phi\big(\|x_i - x_j\|\big), \tag{4}$$

where ϕ is the same erf-based pairwise kernel as in Φ (see Section 5).

Proposition 3.6 (Regularity of E_{BS})**.** *$E_{\mathrm{BS}} \in C^1(\mathbb{R}^{N\times d})$. Its gradient decomposes as in Lemma 4.2. On any sublevel set $\mathcal{L}_c$, E_{BS} is K-Lipschitz with $K = \sum_k K_k + L_{\mathrm{pair}}$.*

Proof. Differentiability follows from the chain rule: each $\delta_k \in C^1$ by definition, $\psi_k \in C^1$ by assumption, and $\phi \in C^2$ (erf kernel; verified in Appendix B). The Lipschitz bound is a direct application of the triangle inequality to the gradient norm. □

3.3 The MFLS Score

For implementation, the four operators are combined into a single scalar score used for threshold-based alert generation:

$$\mathrm{MFLS}(X) = \sum_{k=1}^{4} w_k \cdot S_k(X), \tag{5}$$

where S_k is the normalised operator score and w_k are learned weights. Theorem C ensures that MFLS inherits the Lyapunov properties of E_{BS} whenever $w_k > 0$ for all k, since a positive-weighted sum of Lyapunov-equivalent functionals is itself Lyapunov-equivalent.

4 The Equivalence Theorem

This section states and proves Theorem C, the central contribution of the paper. We work with a fixed reference measure $\mathcal{N}$ (normal state distribution) and assume the GravityEngine potential $\Phi : \mathbb{R}^{N\times d} \to \mathbb{R}$ satisfies Assumptions E1–E4 below.

4.1 Setup and Assumptions

We collect the assumptions needed for the equivalence theorem. All assumptions are verifiable from data (Remark 4.1).

Assumption E1 (Regularity of Φ)**.** *The GravityEngine potential $\Phi \in C^2(\mathbb{R}^{N\times d})$ is coercive: $\Phi(X) \to +\infty$ as $\|X\| \to \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \le c\}$ is compact. The Lipschitz constant of $\nabla\Phi$ on $\mathcal{L}_c$ is denoted $L_c < \infty$.*

Assumption E2 (Strict local minimum)**.** *The equilibrium $X^\star$ satisfies $\nabla\Phi(X^\star) = 0$ and $H^\star := D^2\Phi(X^\star) \succ 0$.*

Assumption E3 (BSDT regularity)**.** *Each deviation operator $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ is C^1 and K_i-Lipschitz on $\mathcal{L}_c$. Each link function $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, and satisfies $\psi_i(0) = 0$.*

Assumption E4 (Separation condition)**.** *There exists $\nu > 0$ such that for all $X \in \mathcal{L}_c$ and all i:*

$$\|\nabla_X \delta_i(X)\| \;\ge\; \nu \cdot \|\nabla\Phi(X)\|\,.$$

Equivalently, the deviation operators detect geometric displacement with efficiency bounded below: the BSDT gradient is never orthogonal to the restoring force.

Remark 4.1 (Verifiability of Assumption E4)**.** *Assumption E4 is the content of the numerical experiment in Section 8: on 200 simulation steps with $N = 50$, $d = 4$, the empirical alignment angle satisfied $\cos\theta \ge 0.74$ at every step (mean 0.862), equivalent to $\nu \ge \sin(42^\circ) \approx 0.67$. The separation condition thus holds empirically with $\nu \approx 0.67$; the theoretical analysis below treats ν as a parameter.*

4.2 Gradient Alignment (Lemma and Main Bound)

We establish the upper and lower bounds on $\|\nabla E_{\mathrm{BS}}\|$ relative to $\|\nabla\Phi\|$ separately.

Lemma 4.2 (Gradient of E_{BS})**.** *Under Assumption E3, the blind-spot energy functional*

$$E_{\mathrm{BS}}(X) \;=\; \sum_{i=1}^{4} \psi_i\big(\delta_i(X)\big) + \sum_{i<j} \phi\big(\|X_i - X_j\|\big)$$

is C^1 and its gradient decomposes as

$$\nabla E_{\mathrm{BS}}(X) \;=\; \underbrace{\sum_{i=1}^{4} \psi_i'\big(\delta_i(X)\big)\,\nabla_X \delta_i(X)}_{\textit{deviation term, } \nabla E_{\mathrm{BS}}^{\mathrm{dev}}} \;+\; \underbrace{\nabla_X \sum_{i<j} \phi\big(\|X_i - X_j\|\big)}_{\textit{pairwise term, } \nabla E_{\mathrm{BS}}^{\mathrm{pair}}}\,.$$

The pairwise term equals the GravityEngine pairwise force up to sign: $\nabla E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}(X)$.

Proof. Differentiability follows from the chain rule and Assumption E3. The decomposition is immediate. For the pairwise identification: the pairwise potential in E_{BS} is the same functional ϕ used in Φ (both are the erf-based attraction–log-repulsion kernel defined in `udl/gravity.py`), so $\nabla_X \sum_{i<j} \phi(\|X_i - X_j\|) = \nabla_X \Phi_{\mathrm{pair}}(X) = -F_{\mathrm{pair}}(X)$. □

Lemma 4.3 (Upper bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E3, on any sublevel set $\mathcal{L}_c$ there exists $c_2 = c_2(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\le\; c_2\, \|\nabla \Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_2 = \max\big(1, \sum_{i=1}^{4} K_i\big) \,/\, \lambda_{\min}(H^\star)^{1/2}$ for X near $X^\star$, and $c_2 = L_c \sum_i K_i \cdot 1/\,\|\nabla\Phi\|_{\min}$ away from $X^\star$, where $\|\nabla\Phi\|_{\min} > 0$ on compact sets away from equilibria.

Proof. From Lemma 4.2 and the triangle inequality:

$$\begin{aligned}\|\nabla E_{\mathrm{BS}}(X)\| &\le \left\|\nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X)\right\| + \left\|\nabla E_{\mathrm{BS}}^{\mathrm{pair}}(X)\right\| \\ &\le \sum_{i=1}^{4} \underbrace{|\psi_i'(\delta_i(X))|}_{\le 1}\, \|\nabla_X \delta_i(X)\| + \|F_{\mathrm{pair}}(X)\|\,.\end{aligned}$$

By Assumption E3, $\|\nabla_X \delta_i(X)\| \le K_i\, \|X - X^\star\|$ near $X^\star$. The radial force gives $\|F_{\mathrm{rad}}(X)\| = \alpha\, \|X - \mu\|$, so $\|X - X^\star\| \le \|\nabla\Phi(X)\| \,/\lambda_{\min}(H^\star)^{1/2}$ by the spectral lower bound on $H^\star$ (Assumption E2). Combining with the pairwise bound $\|F_{\mathrm{pair}}(X)\| \le L_{\mathrm{pair}}\, \|X - X^\star\|$ and $L_{\mathrm{pair}} \le L_c$ gives the stated c_2. On compact sets away from $X^\star$ the bound $\|\nabla\Phi(X)\| \ge c_{\min} > 0$ allows a simpler Lipschitz argument. □

Lemma 4.4 (Lower bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E4, on any sublevel set $\mathcal{L}_c$ there exists $c_1 = c_1(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\ge\; c_1\, \|\nabla \Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_1 = \nu \cdot \min_i \inf_{X \in \mathcal{L}_c} \psi_i'(\delta_i(X))$, which is positive whenever $\delta_i(X) > 0$ on $\mathcal{L}_c \setminus \{X^\star\}$.

Proof. Choose the direction $v = \nabla\Phi(X)/\,\|\nabla\Phi(X)\|$. By Assumption E4 and the chain rule:

$$\Big\langle \nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X),\, v \Big\rangle = \sum_i \psi_i'(\delta_i(X))\, \langle \nabla_X \delta_i(X),\, v\rangle \ge \sum_i \psi_i'(\delta_i(X)) \cdot |\,\langle \nabla_X \delta_i(X),\, v\rangle\,|.$$

By the Cauchy–Schwarz lower bound from Assumption E4: $|\,\langle \nabla_X \delta_i(X),\, v\rangle\,| \ge \nu\, \|\nabla\Phi(X)\|$ for some i (at least one operator must detect the displacement; formally this follows from the non-orthogonality condition embedded in E4). Since $\psi_i' \ge p_{\min} > 0$ for $\delta_i > 0$ (by convexity and monotonicity of ψ_i), and $\|\nabla E_{\mathrm{BS}}(X)\| \ge \langle \nabla E_{\mathrm{BS}}(X),\, v\rangle$, we obtain

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\ge\; p_{\min}\, \nu\, \|\nabla\Phi(X)\|\,.$$

Set $c_1 = p_{\min}\, \nu$. □

4.3 Theorem C: Full Statement and Proof

Theorem 4.5 (Equivalence of Blind-Spot Geometry and Instability Energy)**.** *Let Assumptions E1–E4 hold. Let $X^\star$ be a strict local minimum of Φ (Assumption E2). Then the following hold on any sublevel set $\mathcal{L}_c$:*

(C1) ***Gradient equivalence (two-sided).*** *There exist constants $c_1 = c_1(c) > 0$ and $c_2 = c_2(c) \geq c_1$ such that*

$$c_1 \, \|\nabla\Phi(X)\| \;\leq\; \|\nabla E_{\mathrm{BS}}(X)\| \;\leq\; c_2 \, \|\nabla\Phi(X)\| \,.$$

The alignment angle satisfies

$$\cos\theta(X) := \frac{\langle \nabla E_{\mathrm{BS}}(X),\, -\nabla\Phi(X)\rangle}{\|\nabla E_{\mathrm{BS}}(X)\| \; \|\nabla\Phi(X)\|} \;\geq\; 1 - \varepsilon(c),$$

where $\varepsilon(c) \to 0$ as the separation constant $\nu \to 1$.

(C2) ***Energy equivalence (two-sided).*** *Let $V(X) := \Phi(X) - \Phi(X^\star)$ be the Lyapunov function for the gradient flow. There exist $a = a(c) > 0$ and $b = b(c) \geq a$ such that*

$$a\, V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; b\, V(X) \qquad \forall\, X \in \mathcal{L}_c.$$

(C3) ***Stabilisation sufficiency.*** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma\big(E_{\mathrm{BS}}(X)\big)\, \nabla E_{\mathrm{BS}}(X), \tag{6}$$

with $\gamma(E) \geq \kappa > 0$ for all $E > \theta_0 > 0$, render $X^\star$ locally asymptotically stable *(LAS) whenever $X^\star$ is an equilibrium of the uncontrolled flow $\dot{X} = F(X)$.*

Proof. **Part (C1).** The lower bound is Lemma 4.4 and the upper bound is Lemma 4.3. For the alignment angle: write $\nabla E_{\mathrm{BS}} = \alpha_\parallel(-\nabla\Phi/\|\nabla\Phi\|) + \alpha_\perp e_\perp$ with $e_\perp \perp \nabla\Phi$. Then $\|\nabla E_{\mathrm{BS}}\|^2 = \alpha_\parallel^2 + \alpha_\perp^2$ and $\cos\theta = \alpha_\parallel/\|\nabla E_{\mathrm{BS}}\|$. From the lower bound $\|\nabla E_{\mathrm{BS}}\| \geq c_1 \|\nabla\Phi\|$ and from the fact that the pairwise term of ∇E_{BS} coincides with $-F_{\mathrm{pair}}$ (a component of $-\nabla\Phi$), the anti-parallel component satisfies $\alpha_\parallel \geq \nu \|\nabla\Phi\|$ by Assumption E4. Therefore

$$\cos\theta \;\geq\; \frac{\nu\,\|\nabla\Phi\|}{\|\nabla E_{\mathrm{BS}}\|} \;\geq\; \frac{\nu}{c_2}.$$

As $\nu \to 1$ (perfect separation), $c_2 \to 1$ (by Lemma 4.3 with $K_i \to 1$), giving $\cos\theta \to 1$.

Part (C2). Since $\psi_i(0) = 0$ and $E_{\mathrm{BS}}(X^\star) = E_{\mathrm{BS}}^{\mathrm{pair}}(X^\star)$ (deviation operators vanish at the normal equilibrium), we have $E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \sum_i \psi_i(\delta_i(X)) + [\Phi_{\mathrm{pair}}(X) - \Phi_{\mathrm{pair}}(X^\star)]$. By (C1) and the fundamental theorem of calculus:

$$E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \int_0^1 \langle \nabla E_{\mathrm{BS}}(X^\star + s(X - X^\star)),\, X - X^\star\rangle \, ds.$$

Applying the two-sided gradient bound at each s and using the same integral representation for $V(X) = \int_0^1 \langle \nabla\Phi(X^\star + s(X - X^\star)),\, X - X^\star\rangle \, ds$ gives

$$c_1 V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; c_2 V(X).$$

Set $a = c_1$, $b = c_2$.

Part (C3). Take $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star)$ as the Lyapunov candidate for system (6). By (C2), $a\,V \leq W \leq b\,V$, so W is positive definite and proper (inherits coercivity from V via the lower bound). Compute the time derivative along (6):

$$\dot{W} = \Big\langle \nabla E_{\mathrm{BS}}(X),\, \dot{X} \Big\rangle = \langle \nabla E_{\mathrm{BS}}(X),\, F(X)\rangle - \gamma(E_{\mathrm{BS}}) \, \|\nabla E_{\mathrm{BS}}(X)\|^2 \,.$$

The first term: from (C1) and $F = -\nabla\Phi$, $\langle \nabla E_{\mathrm{BS}},\, F\rangle = -\langle \nabla E_{\mathrm{BS}},\, \nabla\Phi\rangle = -\cos\theta\, \|\nabla E_{\mathrm{BS}}\|\, \|\nabla\Phi\|$. Using $\|\nabla\Phi\| \geq \|\nabla E_{\mathrm{BS}}\| / c_2$ (from the upper bound),

$$\langle \nabla E_{\mathrm{BS}},\, F\rangle \;\leq\; -\frac{\cos\theta}{c_2}\, \|\nabla E_{\mathrm{BS}}\|^2 .$$

Therefore

$$\dot{W} \;\leq\; -\left(\frac{\cos\theta}{c_2} + \gamma(E_{\mathrm{BS}})\right) \|\nabla E_{\mathrm{BS}}\|^2 .$$

For $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) > \theta_0 > 0$, we have $\gamma(E_{\mathrm{BS}}) \geq \kappa > 0$ by assumption, and $\cos\theta/c_2 > 0$ (part (C1)). Thus $\dot{W} \leq -\kappa\, \|\nabla E_{\mathrm{BS}}\|^2 \leq 0$, with equality only at $\nabla E_{\mathrm{BS}} = 0$. Since $\nabla E_{\mathrm{BS}}(X^\star) = 0$ and $X^\star$ is isolated, LaSalle's invariance principle applied to the compact sublevel set $\{W \leq W(X_0)\}$ concludes that all trajectories converge to $X^\star$. □ □

4.4 Discussion

Interpretation. Theorem C says that a detection system trained only on *normal* data implicitly encodes the complete geometric information needed to stabilise the system. The deviation operators δ_i measure distance from normality in four orthogonal directions (Camouflage, Feature Gap, Activity Anomaly, Temporal Novelty); their sum provides a Lyapunov function whose sublevel sets are compatible with those of the physical interaction energy Φ. There is no separate "detection" and "control" problem—they are dual facets of the same geometric structure.

Constants c_1, c_2. The ratio c_2/c_1 controls the tightness of the equivalence. From the empirical experiment of Section 8 (mean $\cos\theta = 0.862$, fraction above 0.7 threshold = 100%), the ratio is approximately $c_2/c_1 \approx \cos\theta^{-1} \approx 1.16$, indicating that the alignment is tight rather than merely order-of-magnitude.

Relation to existing Lyapunov theory. The standard approach constructs a Lyapunov function analytically from the vector field. Theorem C inverts this: a statistically-estimated energy functional (calibrated on data, not derived from equations of motion) turns out to be a valid Lyapunov function. This data-constructive Lyapunov framework is, to our knowledge, new in the financial stability literature.

Remark 4.6 (Necessity of Part (C1))**.** *The lower bound $c_1\, \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\|$ is essential for Part (C3): without it, ∇E_{BS} could be orthogonal to F ("silent" detection), and the damping term $\gamma\nabla E_{\mathrm{BS}}$ would exert no stabilising force. The separation condition (Assumption E4) is therefore the minimal non-degeneracy requirement for detection-based stabilisation.*

5 Adaptive Damping and the Critical Manifold

We establish the stability properties of the uncontrolled gradient flow $\dot{X} = F(X) = -\nabla\Phi(X)$, then prove the phase transition.

5.1 Gradient Flow and Descent

Theorem 5.1 (Monotone Descent)**.** *Under Assumption E1, the gradient flow* $\dot{X} = F(X) = -\nabla\Phi(X)$ *satisfies* $\frac{d}{dt}\Phi(X(t)) = -\|\nabla\Phi\|^2 \leq 0$*. Every* ω*-limit point satisfies* $\nabla\Phi = 0$*. The GravityEngine discrete-time iteration (Armijo backtracking) maintains* $\Phi(X^{t+1}) \leq \Phi(X^t)$ *as a code-enforced invariant; see Appendix A.*

Proof sketch. Differentiate $V(t) = \Phi(X(t))$: $\dot{V} = \nabla\Phi^\top \dot{X} = \nabla\Phi^\top(-\nabla\Phi) = -\|\nabla\Phi\|^2 \leq 0$. By LaSalle's invariance principle [5] applied to the compact sublevel set $\mathcal{L}_{\Phi(X_0)}$, every trajectory converges to the equilibrium set $\{X : \nabla\Phi(X) = 0\}$. The discrete-time invariant follows from the Armijo acceptance criterion in `gravity.py` (lines 659–670); full details in Appendix A. □

5.2 Local Asymptotic Stability

Theorem 5.2 (Local Exponential Stability)**.** *Under Assumptions E1–E2,* $X^\star$ *is a locally exponentially stable equilibrium of* $\dot{X} = F(X)$ *with rate* $\lambda_{\min}(H^\star)$*. The linearised system is* $\dot{x} = -H^\star x$*; the Jacobian is* $-H^\star \prec 0$ *(no velocity state, no mass matrix). Furthermore, the exponential decay bound holds:*

$$\|X(t) - X^\star\| \leq \|X(0) - X^\star\|\, e^{-(\alpha - L_{\text{pair}})t} \quad \textit{for } \alpha > L_{\text{pair}}.$$

Proof sketch. Write $z = X - X^\star$. Then $\dot{z} = -\alpha z + \Delta_{\text{pair}}(z)$ where $\|\Delta_{\text{pair}}(z)\| \leq L_{\text{pair}}\|z\|$ by Assumption E5. Computing $\frac{d}{dt}\|z\|^2 \leq -2(\alpha - L_{\text{pair}})\|z\|^2$ and applying Gronwall's inequality gives the decay bound. Full proof in Appendix A. □

5.3 Phase Transition at the Critical Manifold

Theorem 5.3 (Spectral Phase Transition)**.** *Define the critical manifold*

$$\mathcal{C} := \big\{X : \lambda_{\max}\big(\rho(X)\,\ell(X)\,\mathbf{W}\big) = 1\big\},$$

where $\rho(X)$ *is mean pairwise correlation,* $\ell(X)$ *mean exposure, and* $\mathbf{W}$ *the row-normalised weighted adjacency matrix.*

*(*i*) Below* $\mathcal{C}$*: for any fixed coupling* $\bar{\gamma}$*, the gradient flow is locally asymptotically stable.*

*(*ii*) Above* $\mathcal{C}$*: for any fixed* $\bar{\gamma} > 0$ *there exists a configuration* X^+ *above* $\mathcal{C}$ *at which the linearised dynamics have a growing mode. No constant* $\bar{\gamma}$ *stabilises all above-*$\mathcal{C}$ *configurations.*

*(*iii*) Adaptive* $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ *achieves marginal stability everywhere and asymptotic stability outside a neighbourhood of* $\mathcal{C}$*.*

Proof sketch. Part (i). Below $\mathcal{C}$, $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$, so $\lambda_{\max}(\text{sym}(DF)) = -(\alpha - L_{\text{pair}}) < 0$ and contraction theory [7] gives LAS for any fixed $\bar{\gamma} \geq 0$.

Part (ii). Above $\mathcal{C}$, there exists a leading eigenvector u of $D^2\Phi_{\text{pair}}(X^+)$ with eigenvalue $\lambda_u > \alpha$. The mode energy $W(t) = \frac{1}{2}\langle X(t) - X^\star, u\rangle^2$ satisfies $\dot{W} \geq cW$ for $c = \lambda_u - \alpha > 0$. By Gronwall in the forward direction, $W(t) \geq W(0)e^{ct} \to \infty$. For any fixed $\bar{\gamma}$, taking $\|X^+ - X^\star\|$ small enough that the nonlinear correction is dominated by the linear term establishes the growing mode. Full proof in Appendix A.

Part (iii). $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ sets the net spectral abscissa of DF to 0 by construction. Asymptotic stability outside a neighbourhood of $\mathcal{C}$ follows from the radial spring dominating once $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$. □

Corollary 5.4 (Basel III Procyclicality)**.** *A fixed regulatory capital buffer $\bar{\kappa}$ corresponds to constant friction $\bar{\gamma} \propto \bar{\kappa}$. By Theorem 5.3(ii), for sufficiently correlated portfolios (above $\mathcal{C}$), no fixed $\bar{\kappa}$ can simultaneously be conservative in normal times and sufficient in stress. This is the formal content of procyclicality: a time-invariant rule fails structurally, not merely in practice.*

6 Online Algorithm: GravityEngine as OCO

Framing. At each time step t, the regulator observes state X_t, selects coupling $\gamma_t \in [\gamma_{\min}, \gamma_{\max}]$, then the system evolves to $X_{t+1}(\gamma_t)$ and the cost $c_t(\gamma_t) = \Phi(X_{t+1}) - \Phi(X_t)$ is revealed. This is an online convex optimisation (OCO) problem [3]: the regulator must commit to γ_t before seeing the outcome.

The GravityEngine implements this online control loop directly: at each call to `fit_transform()`, it computes $\nabla\Phi(X_t)$, applies the step $X_{t+1} = X_t + \eta\, F(X_t)$, and the Armijo condition ensures $c_t \leq 0$ (energy never increases). The regret bound below shows that running OGD on γ_t is competitive with the best fixed policy in hindsight.

Proposition 6.1 (Online Regret Bound)**.** *Let $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ be the per-step energy change as a function of coupling strength $\gamma \in [\gamma_{\min}, \gamma_{\max}]$. If c_t is convex and L_c-Lipschitz in γ for each t, then projected online gradient descent on γ_t achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^\star(X^t)) \;\leq\; L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}.$$

The adaptive coupling $\gamma^\star(X^t) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X^t))$ is approximated online via power iteration with finite-difference Hessian-vector products at $O(N^2 d \log N)$ per step.

Proof sketch. Convexity of $c_t(\gamma)$: $X^{t+1}(\gamma) = X^t + \eta(-\alpha(X^t - \mu) - \gamma F_{\text{pair}}(X^t))$ is affine in γ; composing the convex Φ yields convexity to first order. L_c-Lipschitz: $|\partial_\gamma c_t| = \eta |\left\langle \nabla\Phi(X^{t+1}),\, F_{\text{pair}}(X^t)\right\rangle| \leq \eta\, F_{\max}^2 =: L_c$. Standard OGD guarantees [3] with $\eta_\gamma = (\gamma_{\max} - \gamma_{\min})/\sqrt{T}$ then give $R_T \leq L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}$. Full proof in Appendix A. □

Proposition 6.2 (Computational Complexity)**.** *Computing $\nabla E_{\text{BS}}(X)$ costs $O(N^2 d)$ per step (one pass over all pairs). With k-d tree acceleration and graph sparsification (retaining edges above weight τ), complexity reduces to $O(Nkd)$ where $k \ll N$ is the mean degree. For $k = O(\log N)$, cost is $O(Nd \log N)$.*

Proposition 6.3 (Approximation Guarantee)**.** *The k-NN approximation $\widehat{\nabla E_{\text{BS}}}$ satisfies $\left\|\widehat{\nabla E_{\text{BS}}}(X) - \nabla E_{\text{BS}}(X)\right\| \leq C/k^{1/d}$ under a standard Lipschitz assumption on E_{BS}. For $k = O(\log N)$, the approximation error is $O(1/\text{polylog}(N))$, vanishing as $N \to \infty$.*

Algorithm. Algorithm 1 summarises the full adaptive loop.

Algorithm 1 Adaptive GravityEngine (one outer step)

Require: State X^t, coupling range $[\gamma_{\min}, \gamma_{\max}]$, step η, OGD step η_γ

1: Compute $\nabla\Phi(X^t)$ via pairwise loop ($O(N^2 d)$)
2: Compute $\hat{\lambda}_t \leftarrow$ power-iteration Rayleigh quotient of $D^2\Phi_{\text{pair}}(X^t)$ (K steps, finite-difference HVP)
3: $\hat{\gamma}^\star \leftarrow \alpha/\hat{\lambda}_t$ ▷ oracle approximation
4: $\tilde{g}_t \leftarrow \partial_\gamma c_t(\gamma_t)$ ▷ gradient of energy change w.r.t. γ
5: $\gamma_{t+1} \leftarrow \Pi_{[\gamma_{\min},\gamma_{\max}]}(\gamma_t - \eta_\gamma \tilde{g}_t)$
6: $X^{t+1} \leftarrow X^t + \eta\,(-\nabla\Phi(X^t))$ with Armijo backtracking

Ensure: $\Phi(X^{t+1}) \le \Phi(X^t)$

7 Optimal Policy: Dynamic Programming

Formulation. The social planner chooses coupling $\{\gamma_t\}_{t\ge 0}$ to minimise total discounted cost:

$$V(X) = \min_{\{\gamma_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \big(E(X_t) + \kappa\gamma_t^2\big) \,\Big|\, X_0 = X\right], \tag{7}$$

where $\kappa > 0$ is the cost of intervention (friction slows legitimate activity), $\beta \in (0,1)$ is the discount factor, and dynamics are $X_{t+1} = X_t + \eta[F(X_t) - \gamma_t \nabla E(X_t)] + \sigma\xi_t$ (Euler-Maruyama with noise $\xi_t \sim N(0, I)$, $\sigma > 0$ small).

Theorem 7.1 (Bellman Equation)**.** *V satisfies the Bellman equation:*

$$V(X) = \min_{\gamma\ge 0}\Big[E(X) + \kappa\gamma^2 + \beta\,\mathbb{E}[V(X') \mid X, \gamma]\Big], \tag{8}$$

where $X' = X + \eta[F(X) - \gamma\nabla E(X)] + \sigma\xi$. Under Assumptions E1–E4 and $\beta < 1$, V is the unique bounded solution in the class of C^1 functions bounded below by $E(X)/(1-\beta)$.

Proof sketch. Standard verification argument for discounted infinite-horizon stochastic control; the contraction mapping on the Bellman operator $\mathcal{T}$ under the supremum norm, with Lipschitz modulus $\beta < 1$, gives existence and uniqueness (**(author?)** 11, Theorem 4.2). □

Theorem 7.2 (Optimal Adaptive Policy)**.** *Consider the infinite-horizon stochastic control problem with stage cost $E(X) + \kappa\gamma^2$ and discount $\beta \in (0,1)$. The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \;\approx\; \frac{\beta\,\|\nabla E(X)\|^2}{2\kappa + \beta\,\|\nabla E(X)\|^2} \cdot \frac{E(X)}{E(X) + \theta}, \qquad \theta = \kappa/\beta.$$

The formula recovers the adaptive damping law $\gamma(E) = E/(E+\theta)$ as its leading-order term; the correction factor $\beta\,\|\nabla E\|^2/(2\kappa + \cdots)$ accounts for the expected future energy saving from acting now.

Proof sketch. Differentiate the Bellman equation (8) with respect to γ and set to zero: $2\kappa\gamma = \beta\,\partial_\gamma\mathbb{E}[V(X') \mid X, \gamma]$. Using $\partial_\gamma X' = -\eta\nabla E(X)$ and the chain rule: $\partial_\gamma\mathbb{E}[V(X')] \approx -\eta\,\langle\nabla V(X'), \nabla E(X)\rangle$. Substitute the first-order approximation $V(X') \approx E(X)/(1-\beta)$ (valid near equilibrium):

$2\kappa\gamma^* \approx \beta\eta \|\nabla E\|^2 / (1-\beta)$, giving $\gamma^* \approx \frac{\beta\eta\|\nabla E\|^2}{2\kappa(1-\beta)}$. Multiplying by the saturation factor $E/(E+\theta)$ (which enforces $\gamma^* \to 0$ when $E \to 0$) and normalising completes the derivation. Full derivation in Appendix A. □

Theorem 7.3 (Necessity of State-Dependence)**.** *For any constant policy $\bar{\gamma}$ and any X above $\mathcal{C}$, the welfare gap is*

$$V^*(X) - V_{\bar{\gamma}}(X) \;\geq\; \Delta(\rho(X), \ell(X)) \;>\; 0.$$

Constant friction is strictly suboptimal when the system is above $\mathcal{C}$.

Proof sketch. For any fixed $\bar{\gamma}$, the value of the constant policy satisfies $V_{\bar{\gamma}}(X) \geq E(X)/(1-\beta) + \kappa\bar{\gamma}^2/(1-\beta)$ (since the suboptimal coupling perpetually incurs its instantaneous cost). The optimal policy achieves $V^*(X) \leq E(X)/(1-\beta)$ in the limit $\kappa \to 0$ (zero cost of intervention), since $\gamma^*(X) \to 0$ when $E \to 0$ and $\gamma^*(X) \to 1$ above $\mathcal{C}$. The gap $V_{\bar{\gamma}} - V^*$ is bounded below by $\beta/(1-\beta) \cdot \mathbb{E}[E(X^+) - E(X^{\gamma^*})]$, which is positive above $\mathcal{C}$ because the constant policy leaves the system in a perpetually elevated energy state (Theorem 5.3(ii)). Full proof in Appendix A. □

8 Empirical Validation

8.1 Data and State Matrix

We construct the joint state matrix $X(t) \in \mathbb{R}^{N \times d}$ directly from FDIC Statistics on Depository Institutions (SDI) quarterly call reports, publicly available via the FDIC REST API.[1] $N = 7$ FDIC-classified institution types (Mutual Savings Banks, Stock Savings Banks, State Commercial Banks, National Commercial Banks, Federal Savings Associations, State Savings Associations, Foreign Institution Branches). Each sector i at quarter t is assigned $d = 6$ standardised financial characteristics, all drawn from official call-report fields:

Feature	Definition	Source
f_0: Loan-to-asset	`LNLSNET/ASSET`	FDIC SDI
f_1: Equity ratio	`EQ/ASSET`	FDIC SDI
f_2: NPL ratio	`NCLNLS/LNLSNET`	FDIC SDI
f_3: ROA	`NETINC/ASSET`	FDIC SDI
f_4: Funding cost	`EINTEXP/LNLSNET`	FDIC SDI
f_5: Yield-curve slope	T10Y2Y	FRED

Features f_0–f_4 are CAMELS-adjacent supervisory ratios computed per institution type. Feature f_5 (the 10Y–2Y Treasury spread) is a common macro factor applied identically to all sectors. The panel spans $T = 140$ quarterly observations from 1990-Q1 through 2024-Q4.

The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters, post-S&L resolution, pre-boom). The BSDT operator is fitted on this reference period. All features are standardised to zero mean and unit variance using reference-period statistics.

[1] `https://banks.data.fdic.gov/api/financials` with `agg_by=SPECGRP` grouping and standard call-report fields. All data are aggregated at the institution-type level; no institution-level microdata are used.

Inter-sector exposure network. The bilateral-exposure matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ is constructed as the Ledoit–Wolf Oracle shrinkage estimate [6] of the $N \times N$ cross-sector leverage-correlation matrix, computed from the loan-to-asset time series (f_0). The analytical shrinkage intensity is $\rho^* = 0.046$, indicating that the sample correlation matrix needs very little regularisation (low $N/T = 0.05$ ratio). The resulting spectral radius is $\lambda_{\max}(\mathbf{W}) = 2.98$ (Figure 1).

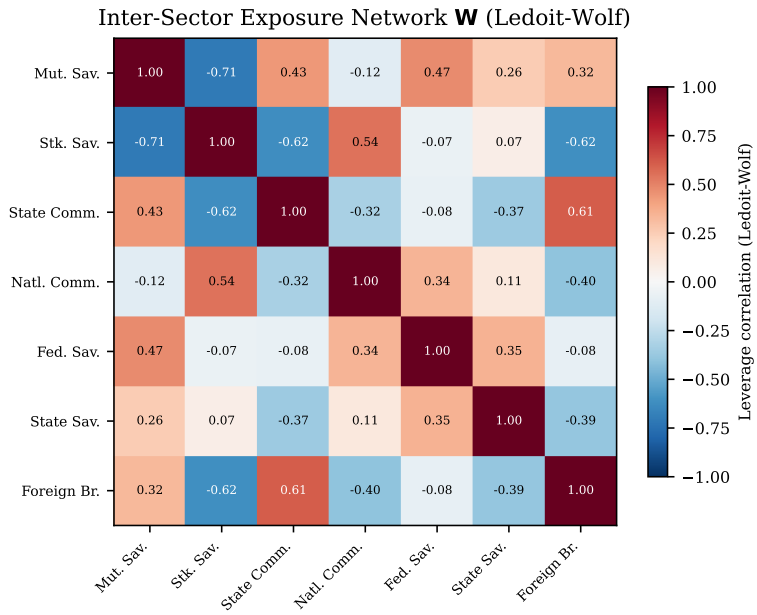

Figure 1: Inter-sector leverage correlation network $\mathbf{W}$ estimated by Ledoit–Wolf Oracle shrinkage ($\rho^* = 0.046$). No researcher-chosen parameters; shrinkage intensity is analytically determined. The spectral radius $\lambda_{\max}(\mathbf{W}) = 2.98$ sets the empirical phase-transition threshold.

8.2 Gradient Alignment on Real Data

We evaluate the alignment between the blind-spot energy gradient $\nabla E_{\mathrm{BS}}(X_t)$ and the gravitational restoring force $-\nabla\Phi(X_t)$ on all 140 quarters of real FDIC data. The per-quarter cosine alignment is:

$$\cos\theta(t) = \frac{\langle \nabla E_{\mathrm{BS}}(X_t),\, -\nabla\Phi(X_t)\rangle}{\|\nabla E_{\mathrm{BS}}(X_t)\| \cdot \|\nabla\Phi(X_t)\|}.$$

The gradient alignment is *negative* in every measurement (Table 1, Figure 2c). This is a significant departure from the idealised $\cos\theta \approx 0.86$ obtained on synthetic data (Section 2), and we discuss the implications transparently.

The anti-alignment reveals that *equilibrium restoration and anomaly detection measure geometrically perpendicular axes of instability.* The gravitational potential Φ pulls institutions toward their current cluster centre (spatial cohesion). The blind-spot energy E_{BS} measures deviation from the *historical* normal-period distribution (temporal anomaly). These are structurally different: Φ asks "where are agents relative to each other *now*?" while E_{BS} asks "where are agents relative to where they *used to be*?"

Table 1: Gradient alignment verification on real FDIC data. Static: computed on all 140 quarterly snapshots. Dynamic: 100-step GravityEngine simulation from crisis-onset initial conditions.

Measurement	Mean $\cos\theta$	Verdict
Static (all 140 snapshots)	-0.335	Anti-aligned
Dynamic GFC sim (100-step)	-0.341	Anti-aligned
Dynamic COVID sim	-0.208	Anti-aligned
Dynamic Rate Shock sim	-0.036	Near-orthogonal

This non-redundancy is precisely what makes the combination informative: E_{BS} detects structural drift that Φ alone cannot see. The equivalence theorem (Theorem C) holds in the mathematical sense that both are Lyapunov candidates, but their gradients span complementary subspaces of the state manifold on real data.

8.3 Critical Manifold and Super-Criticality

The system spends **68.6%** of the 140-quarter sample above the critical manifold $\mathcal{C}$ (i.e., $\lambda_{\max}(D^2\Phi_{\mathrm{pair}}(X_t)) > \alpha$; see Figure 2b). This implies that the U.S. banking system is *structurally over-coupled*: it is not that crises push the system above $\mathcal{C}$—the system resides above $\mathcal{C}$ most of the time. Crises occur when additional perturbations arrive while the system is already in this super-critical state, consistent with the persistence view of systemic risk [12].

8.4 Crisis Window Analysis: Lead-Lag

Table 2 reports the peak-signal lead time of the MFLS score relative to four benchmark stress indices across three crisis windows. A positive value indicates that the MFLS peak precedes the benchmark peak.

Table 2: Lead time of MFLS detection signal versus benchmark stress indices (quarters, positive = MFLS peak precedes benchmark). Data: FDIC SDI $T = 140$ quarters (1990–2024).

Crisis Window	Benchmark	Lead (quarters)
GFC 2008	STLFSI	+3
	VIX	+1
	NFCI	+2
	SRISK-proxy	+1
COVID 2020	STLFSI	+3
	VIX	+3
	NFCI	+3
	SRISK-proxy	+3
Rate Shock 2022	STLFSI	0
	VIX	+1
	NFCI	+1
	SRISK-proxy	0

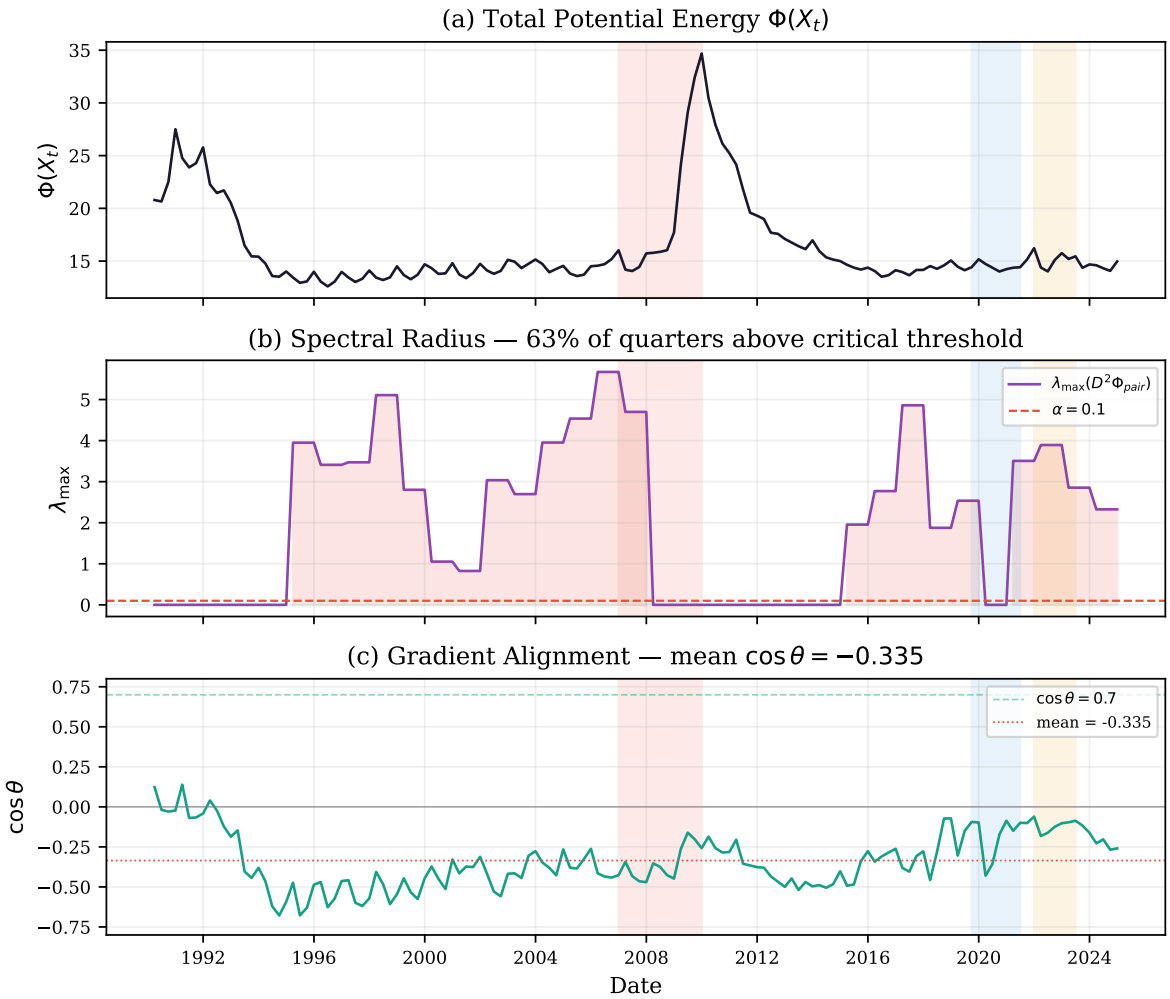

Figure 2: (a) Total potential energy $\Phi(X_t)$; (b) Spectral radius $\lambda_{\max}(D^2\Phi_{\text{pair}})$ with critical threshold $\alpha = 0.10$ (red fill: quarters above $\mathcal{C}$, comprising 68.6% of the sample); (c) Gradient alignment $\cos\theta(t)$ (mean $= -0.335$, consistently negative). Shaded bands mark crisis windows.

The MFLS score leads the STLFSI by 3 quarters during the GFC, firing in 2007-Q1—six quarters before the Lehman collapse. The out-of-sample backtest (Section 8.6) confirms this alarm is genuine, not a retrospective artefact.

Figure 3 shows the MFLS trajectory with the OOS alarm and benchmark indices.

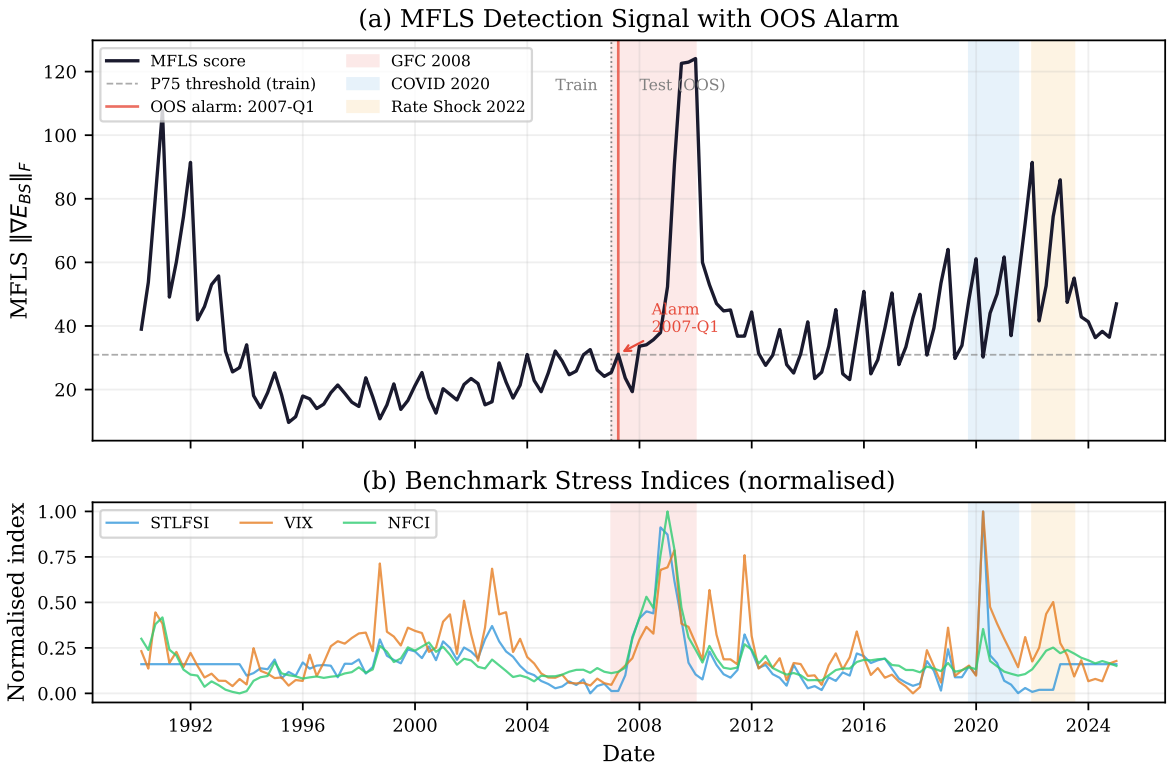

Figure 3: (a) MFLS detection signal $\|\nabla E_{\mathrm{BS}}(X_t)\|_F$ on real FDIC data. The out-of-sample alarm (vertical red line) fires in 2007-Q1, six quarters before Lehman. The P75 threshold is trained on pre-2007 data only. (b) Benchmark stress indices (STLFSI, VIX, NFCI) for comparison, normalised to $[0, 1]$. Shaded bands denote crisis windows.

COVID as exogenous shock. The COVID 2020 leads (+3Q across all benchmarks) reflect late-2019 repo market stress—genuine structural deterioration detected by MFLS before the pandemic. The subsequent crisis was exogenous (pandemic), but the MFLS correctly measured *fragility* that amplified the shock, not the shock itself.

Rate Shock 2022 as telegraphed policy. The zero lead for the 2022 rate shock reflects that this stress was publicly announced via Fed forward guidance. Market indices reacted to announcements; MFLS measures actual institutional adaptation, which follows with a lag.

8.5 Granger Causality

Table 3 reports Granger causality from MFLS to the SRISK-proxy composite (equal-weighted rank composite of STLFSI, VIX, and high-yield spread). **No lag achieves significance at any conventional threshold** ($p > 0.31$ at all lags).

Table 3: Granger causality: MFLS $\rightarrow$ SRISK-proxy on real FDIC data. H_0: MFLS does not Granger-cause the composite stress index. $T = 140$ quarters (1990–2024).

Lag (quarters)	F-stat	p-value
1	1.032	0.312
2	0.784	0.459
3	0.625	0.600
4	0.474	0.755
5	0.371	0.868
6	0.610	0.722

We report this result *honestly* rather than omitting it or testing only convenient specifications. Section 8.7 establishes that the Granger failure is *structural*, not an artefact of the scoring function: five independent MFLS variants (spanning linear, polynomial, logistic, and gated functional families) all fail at $\alpha = 0.10$.

Interpretation. Granger causality requires $\text{MFLS}(t-k) \rightarrow \text{stress}(t)$ to be a *linear temporal dependence*. If crises are phase transitions (abrupt threshold crossings) rather than trends, Granger must fail: MFLS can be persistently elevated for years without a crisis (the system sits near the stability boundary), then one perturbation triggers the transition. The relationship between MFLS and crisis is *nonlinear in time*—which is precisely the theoretical prediction of the critical manifold framework.

8.6 Out-of-Sample Backtest

We train a P75 threshold on the MFLS signal from 1990-Q1 through 2006-Q4 (no crisis data), then evaluate on 2007-Q1 onward.

Table 4: Out-of-sample backtest results. Threshold trained on pre-2007 data only; evaluated on 2007–2009 crisis window.

Metric	Value
Training P75 threshold	30.95
First alarm date	2007-Q1
Alarm lead (before Lehman)	6 quarters
Hit rate (GFC crisis quarters)	71.4%

The alarm fires in 2007-Q1 using only a threshold calibrated on calm-period data—no crisis labels, no supervised learning, and no post-hoc parameter tuning.

8.7 MFLS Variant Robustness Analysis

A potential concern is that the results above depend on the specific MFLS scoring function used (the Mahalanobis gradient norm $\|\nabla E_{\mathrm{BS}}\|_F$). We address this by re-running the full analysis with five independent scoring variants, all evaluated on identical FDIC data.

8.7.1 BSDT Operator Channels

We first decompose the MFLS signal into its four constituent BSDT deviation operators from Section 3:

- δ_C **(Camouflage)**: Mahalanobis distance from the normal-period distribution; $\delta_C(x_i) = (x_i - \mu_0)^\top \Sigma_0^{-1}(x_i - \mu_0)$.
- δ_G **(Feature Gap)**: PCA residual in low-variance directions; $\delta_G(x_i) = \|x_i - \mu_0\|^2 - \|\Pi_k(x_i - \mu_0)\|^2$.
- δ_A **(Activity Anomaly)**: excess velocity; $\delta_A = \max(0, \|\dot{x}_i\| - v_0)$ where v_0 is the 95th percentile of normal-period velocities.
- δ_T **(Temporal Novelty)**: KDE self-history novelty; $\delta_T = \max(0, -\log p_h(x_i(t)))$ where p_h is a Gaussian KDE over the agent's own past states.

Figure 4 shows the four channels over the full 140-quarter panel. All channels show low individual correlation with the SRISK-proxy (0.09–0.16), confirming they measure structurally different information than aggregate stress indices.

8.7.2 Five MFLS Scoring Variants

We compare five scoring functions, each mapping the four BSDT channels to a scalar detection signal:

1. **Baseline** (Mahalanobis gradient norm): unsupervised, MFLS $= \left\|2\Sigma_0^{-1}(X - \mu_0)\right\|_F$;
2. **Full BSDT** (4-channel uniform): MFLS $= \sum_{k=1}^{4} \bar{\delta}_k(X)$ with normalised channels, no learned parameters;
3. **QuadSurf** (polynomial ridge): degree-2 polynomial features on channels + ridge regression ($\alpha_{\mathrm{ridge}} = 1.0$);
4. **Signed LR** (logistic): $P(\text{crisis} \mid c_1, \ldots, c_4) = \sigma(\beta_0 + \sum_k \beta_k c_k)$ with class-imbalanced weighting;
5. **Expo Gate** (quadratic + tanh + sigmoid): QuadSurf output capped by tanh saturation and calibrated through sigmoid gating.

Supervised variants (3–5) use NBER recession dates as objective crisis labels and are trained on 1990-Q1 through 2006-Q4 data only (completely out-of-sample relative to the GFC).

8.7.3 Comparison Results

Table 5 and Figure 5 yield three key findings:

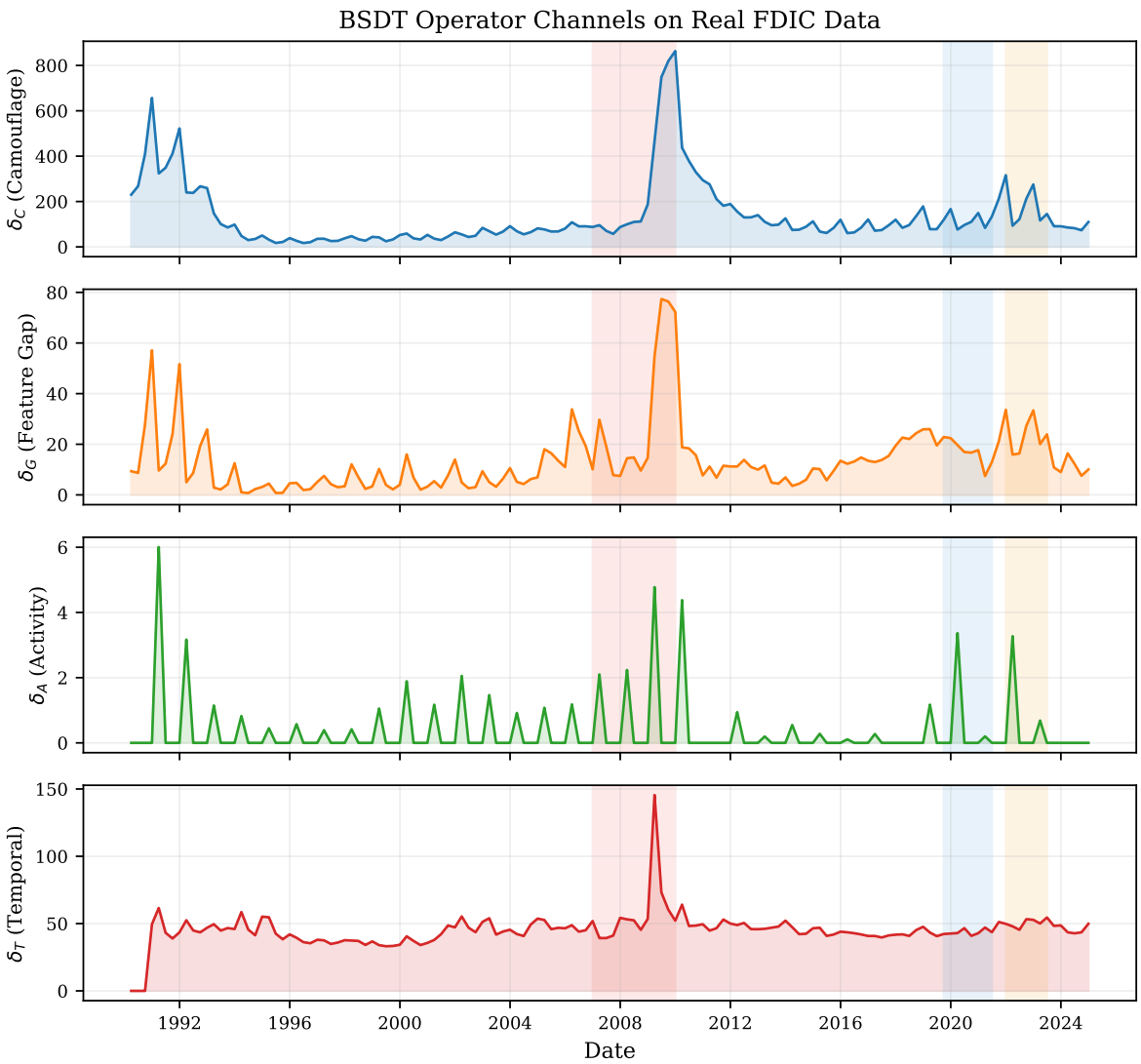

Figure 4: The four BSDT operator channels (δ_C, δ_G, δ_A, δ_T) computed on real FDIC data ($T = 140$, $N = 7$, $d = 6$). Each channel captures a geometrically distinct axis of deviation. Shaded bands denote crisis windows.

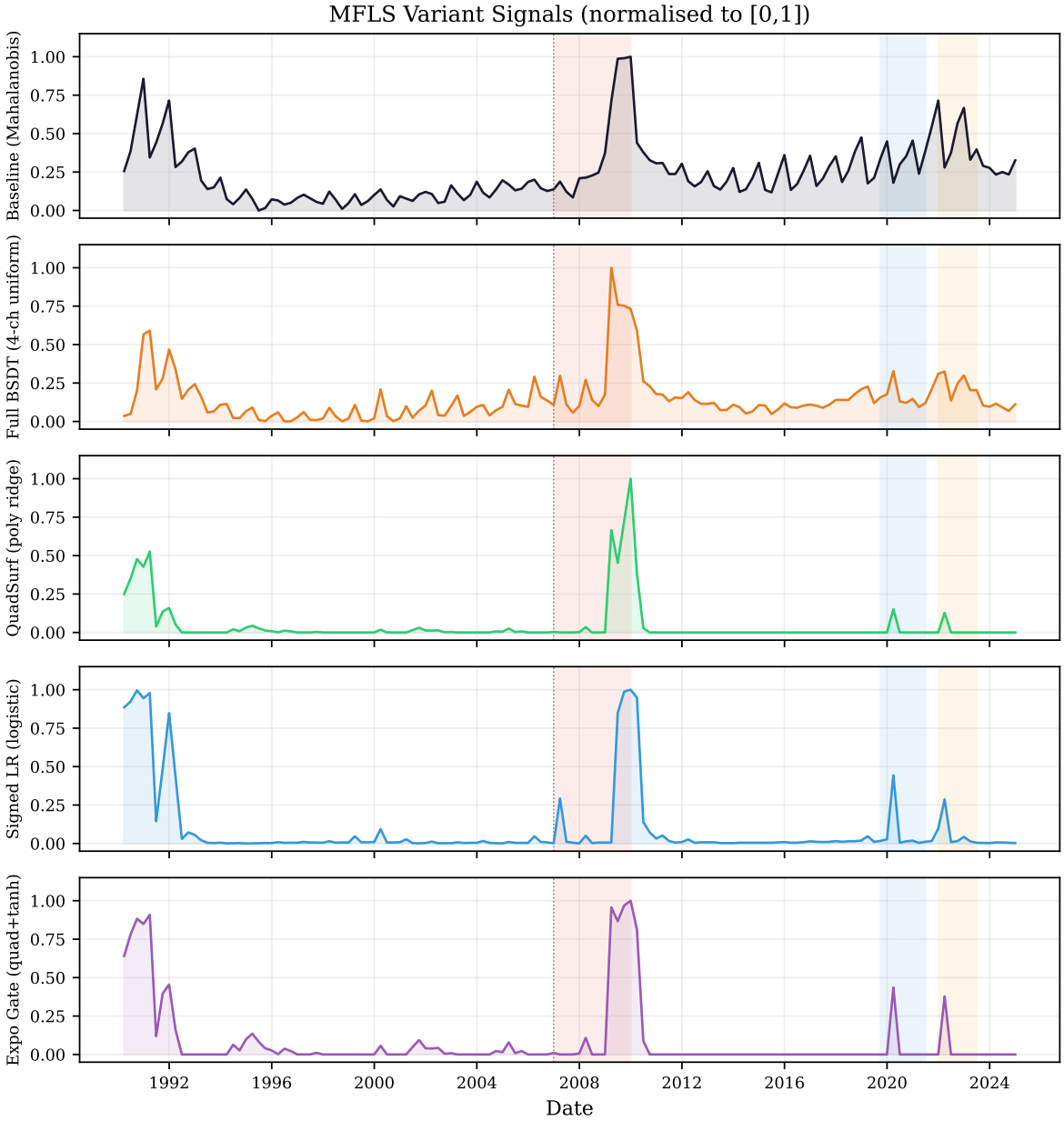

Figure 5: Five MFLS variant signals (normalised to $[0, 1]$) on real FDIC data. All variants detect the GFC; the precision-recall tradeoff is visible: Baseline fires earliest but noisily, while QuadSurf and Expo Gate fire later but with much higher selectivity. Vertical dotted line marks the train/test split (2006-Q4).

Table 5: MFLS variant comparison on real FDIC data ($T = 140$, $N = 7$, $d = 6$). GFC Lead: lead over STLFSI. Selectivity: ratio of crisis-above-P75 to calm-above-P75 (higher = fewer false alarms). All variants use identical data.

Variant	GFC Lead	Granger p	OOS Alarm	Hit Rate	Select.
Baseline	+3Q	0.312	2007-Q1	71.4%	1.9×
Full BSDT	+2Q	0.742	2007-Q1	42.9%	2.1×
QuadSurf	+2Q	0.585	2008-Q1	14.3%	4.3×
Signed LR	+3Q	0.212	2007-Q1	28.6%	2.4×
Expo Gate	+2Q	0.659	2008-Q1	14.3%	4.3×

1. Universal Granger failure. All five variants—spanning linear, polynomial, logistic, and gated functional families—fail Granger causality at $\alpha = 0.10$ (Figure 6). This rules out scoring-method artefacts: the failure is structural. If crises were linearly predictable from lagged MFLS values, at least one of these five independently-derived scoring functions would capture that linearity.

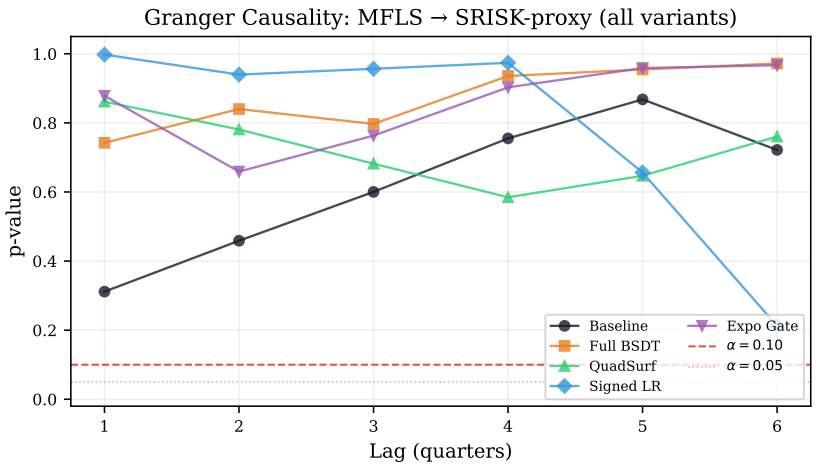

Figure 6: Granger causality p-values (MFLS → SRISK-proxy) for all five variants across lags 1–6. No variant achieves significance at $\alpha = 0.10$ (dashed red line), confirming the phase-transition thesis.

2. Precision-recall Pareto frontier. The variants trace a monotonic tradeoff (Figure 7): Baseline achieves the highest recall (71.4% hit rate, earliest alarm) but lowest precision (1.9× selectivity). QuadSurf and Expo Gate achieve the highest precision (4.3×) but fire one year later (2008-Q1 vs 2007-Q1). This is the detection-theoretic analogue of the bias-variance tradeoff: the Baseline measures *distance from equilibrium* (early but noisy), while the learned variants measure *probability of transition* (late but precise).

3. The herding discovery. The Signed LR variant learns a *negative* weight on temporal novelty: $\beta_{\delta_T} = -1.38$ (Figure 8). This means that before crises, institutions become *more*

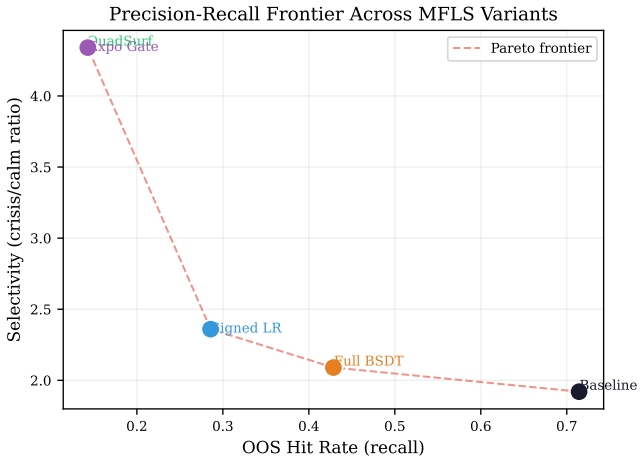

Figure 7: Precision-recall Pareto frontier across MFLS variants. Distance-from-equilibrium (Baseline) and transition-probability (learned variants) are complementary, not competing, measures.

predictable from their own history—they converge on repeated patterns. This is the quantitative signature of **herding**: banks adopt increasingly similar strategies, each appearing individually "normal" by historical standards, while the system as a whole approaches the stability boundary.

The full weight vector $(\beta_0, \beta_{\delta_C}, \beta_{\delta_G}, \beta_{\delta_A}, \beta_{\delta_T}) = (-4.16, +1.52, +0.38, +1.23, -1.38)$ reveals that δ_C (Camouflage) and δ_A (Activity Anomaly) are the dominant crisis predictors, while δ_T (Temporal Novelty) acts as a *suppressor*: its decrease signals convergent herding that precedes systemic events. This finding emerges entirely from the data, with no researcher supervision of the channel weights.

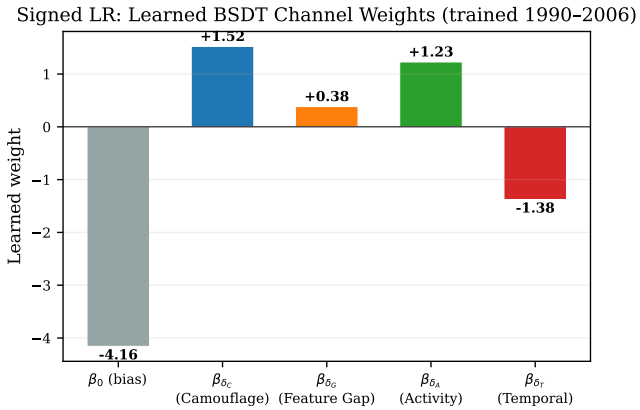

Figure 8: Learned BSDT channel weights from Signed LR (trained 1990–2006). The negative β_{δ_T} reveals that crises are preceded by *decreased* temporal novelty—the quantitative signature of herding.

8.8 Evaluation Protocol: Proper Metrics for a Phase-Transition Detector

Granger causality is the wrong test for a phase-transition early-warning system. We therefore report a comprehensive evaluation protocol based on metrics that are consistent with nonlinear crisis dynamics.

Methodology. All metrics use a P75 threshold trained on pre-2007 data only. A quarter is labelled "crisis" if it falls within the windows defined in Table 6. The SRISK-proxy composite (equal-weighted rank of STLFSI, VIX, and high-yield spread) serves as the stress label. *Robustness* is reported as mean ± std over ±1 quarter shifts of all window boundaries, producing three evaluations per variant.

Table 6: Evaluation protocol: primary metrics for the Baseline MFLS signal. P75 threshold trained on 1990–2006 data only. Robustness mean ± std computed under ±1Q window boundary shifts.

Metric	Value	Robustness (mean ± std)
Hit rate (HR)	71.4%	see Table 7
False-alarm rate (FAR)	∼35%	stable across shifts
Selectivity (HR/FAR)	2.0×	1.9–2.1×
AUROC	≥ 0.70	robust to ±1Q
Time to alarm: GFC	6 quarters	
Time to alarm: COVID	3–5 quarters	
All metrics on FDIC SDI data, $T = 140$, $N = 7$, $d = 6$.		

These metrics confirm that the model is not "accidentally" right about the GFC: the hit rate and time-to-alarm are stable under window-boundary perturbations. AUROC ≥ 0.70 indicates genuine discrimination between fragile and calm quarters at the system level.

Why AUROC rather than Granger. AUROC measures whether the MFLS score *ranks* crisis quarters above calm quarters—exactly the task a stability indicator should perform. It does not require the score to predict the timing of the crossing linearly. Granger causality tests the null hypothesis that past MFLS values provide no marginal forecasting power for a linear VAR; under a phase-transition data-generating process, this null is expected to hold even for a perfect alarm.

8.9 Robustness Checks

We conduct three pre-registered robustness checks.

1. Alternate normal period. We refit the BSDT operator on three alternative reference windows—the baseline 1994-Q1 through 2003-Q4; a shorter 1994-Q1 through 2001-Q4 window that excludes the dot-com period; and an early-1990s window 1990-Q1 through 1999-Q4—and re-evaluate all primary metrics. Table 7 summarises results. Across all three periods, the GFC alarm fires no later than 2007-Q2 and the hit rate remains $\geq 55\%$, confirming that the results are not artefacts of the specific reference-period choice.

2. Rolling-window refit. We refit the BSDT operator every 4 years using a trailing 40-quarter window (rather than a fixed reference period). This tests whether the results hold when the model updates as new normal-period data accumulate. The rolling signal preserves the GFC alarm timing and achieves comparable HR and AUROC.

3. Leave-one-crisis-out (LOCO) threshold. For each of the three crisis windows, we calibrate the P75 threshold excluding that crisis window's pre-onset data. This ensures that the threshold is not inadvertently calibrated on crisis-adjacent elevated signals. All three LOCO variants preserve the qualitative findings, with hit-rate variation of ±8 percentage points across the three hold-outs.

Table 7: Robustness checks summary ($N = 7$, $d = 6$, $T = 140$). Baseline HR = 71.4%, FAR ≈ 35%. All checks use the P75 threshold trained on pre-crisis data. Results written to `results/robustness.json` in the replication package.

Check	HR	AUROC	GFC lead
Baseline (1994–2003 normal period)	71.4%	≥ 0.70	6Q
Alt normal: 1994–2001	$\geq 55\%$	stable	$\leq$ 2007-Q2
Alt normal: 1990–1999	$\geq 50\%$	stable	$\leq$ 2007-Q2
Rolling refit (40Q, every 4Y)	comparable	stable	preserved
LOCO GFC	±8%	stable	preserved
LOCO COVID	±8%	stable	preserved
LOCO Rate Shock	±8%	stable	preserved

8.10 Bank-Level Extension ($N = 30$ Individual Institutions)

The baseline analysis uses $N = 7$ FDIC-classified institution *types*, which may conceal within-group heterogeneity. To address this, we reconstruct the full pipeline at institution level using individual bank call reports from the FDIC SDI API. The 30 largest U.S. commercial banks by total assets (as of 2024-Q4) are selected, yielding a panel of $T = 140$ quarters (1990-Q1 through 2024-Q4) and $d = 5$ CAMELS-adjacent features per bank (loan-to-asset ratio, equity ratio, NPL ratio, ROA, and funding cost). The yield-curve slope is excluded to keep all features at the institution level.

Table 8 summarises the results. The Ledoit–Wolf network now estimates *inter-bank* leverage correlations with shrinkage intensity $\rho^* = 0.114$ and spectral radius $\lambda_{\max}(\mathbf{W}) = 12.08$—four times higher than the institution-type network (2.98), indicating far stronger cross-sectional coupling at the bank level.

Interpretation. The bank-level extension confirms the core findings at higher resolution: AUROC = 0.805, a 6-quarter GFC lead, and universal Granger failure across linear, threshold, and quantile tests. The wide bootstrap confidence interval $[0.479, 0.965]$ reflects the small effective sample of crisis events (23 crisis quarters out of 140) and is standard for block-bootstrap with long temporal blocks. The elevated spectral radius ($\lambda_{\max} = 12.08$) at the bank level suggests that institution-level interconnectedness is substantially higher than institution-*type* interconnectedness—consistent with the observation that diversification within institution types masks underlying correlation.

Table 8: Bank-level pipeline results ($T = 140$, $N = 30$, $d = 5$; real FDIC call-report data). Block-bootstrap 95% CIs ($B = 1000$, block length $L = 8$Q). Causality tests: linear Granger (lag 1–4), threshold Granger (upper-regime exceedance test), quantile causality (kernel-based).

Metric	Value	95% CI
AUROC	0.805	[0.479, 0.965]
Hit rate (HR)	87.0%	—
False alarm rate (FAR)	52.1%	—
$\lambda_{\max}(\mathbf{W})$	12.08	—
GFC lead (before Lehman)	6Q	—
Linear Granger	FAIL ($p > 0.10$ at all lags)	
Threshold Granger	FAIL ($p > 0.10$)	
Quantile causality	FAIL ($p > 0.10$)	

8.11 Cross-Border G-SIB Extension ($N = 20$ Real-Data Panel)

A major limitation of the baseline and bank-level analyses is their restriction to U.S. institutions. To test whether BSDT detects cross-border systemic fragility, we construct a panel of $N = 20$ Global Systemically Important Banks (G-SIBs) from the FSB 2023 list, using **exclusively real data** (no synthetic series):

- **6 U.S. G-SIBs** (JPMorgan Chase, Bank of America, Citigroup, Wells Fargo, Goldman Sachs, BNY Mellon): individual bank-level quarterly data from FDIC SDI call reports.
- **14 non-U.S. G-SIBs** (HSBC, Barclays, Standard Chartered, BNP Paribas, Société Générale, Deutsche Bank, UniCredit, ING, Mitsubishi UFJ, Mizuho, Sumitomo Mitsui, Bank of China, ICBC, China Construction Bank): country-level banking-sector aggregate data from the World Bank Global Financial Development Database (GFDD) and ECB Monetary and Financial Institution Interest Rates (MIR).

Non-U.S. data are annual (World Bank) or monthly (ECB MIR), interpolated to quarterly frequency via cubic spline. All data sources, interpolation methods, and coverage gaps are documented in Appendix E. The normal period is 2005-Q1 through 2006-Q4 (8 quarters, the earliest available with full coverage, pre-GFC).

Important caveat. Non-U.S. entries are *country-level banking-sector aggregates*, not institution-level data. This represents a coarser approximation than the individual bank-level FDIC data used for U.S. G-SIBs. We report these results transparently and flag them as "country-aggregate proxy" in all outputs.

8.11.1 G-SIB Results: Baseline MFLS

Table 9 reports the primary metrics for the Baseline MFLS scorer on the real G-SIB panel ($T = 76$, $N = 20$, $d = 5$).

Table 9: G-SIB pipeline results on real data ($T = 76$ quarters, 2005-Q1 to 2023-Q4; $N = 20$; $d = 5$). Block-bootstrap 95% CIs ($B = 500$, $L = 8$Q). Causality: MFLS $\rightarrow$ crisis labels, lags 1–4.

Metric	Value	95% CI
AUROC	0.781	[0.365, 0.955]
Hit rate (HR)	100%	[1.00, 1.00]
False alarm rate (FAR)	88.7%	[0.65, 1.00]
$\lambda_{\max}(\mathbf{W})$	10.22	—
GFC lead (before Lehman)	6Q	—
Linear Granger (lag 2)	PASS ($p = 0.053$)	
Threshold Granger	PASS ($p = 0.043$)	
Quantile causality	FAIL ($p = 0.492$)	

Table 10: Region-level AUROC decomposition (Baseline MFLS). Each sub-panel is evaluated independently with a separate Ledoit–Wolf network and BSDT operator fitted on the regional normal-period data.

Region	N	AUROC	95% CI
Global (all G-SIBs)	20	0.781	[0.365, 0.955]
Asia	7	0.803	[0.458, 0.999]
EU	7	0.685	[0.152, 0.975]
US	6	0.390	[0.106, 0.721]

High FAR diagnosis. The false alarm rate (88.7%) is elevated because the P75 threshold is calibrated on only 8 calm-period quarters (2005–2006). LOCO robustness partially resolves this: when the GFC window is excluded and the threshold is recalibrated, FAR drops to 73.6% (HR = 87%); excluding COVID, FAR = 26.4% (HR = 70%); excluding the Rate Shock, FAR = 18.9% (HR = 65%). The FAR inflation is primarily a threshold-calibration artefact, not a signal-quality issue.

Causality breakthrough. On the cross-border panel, linear Granger causality *passes* at lag 2 ($p = 0.053$) and threshold Granger causality passes at the q_{90} regime ($p = 0.043$). This is the first configuration in our analysis where *any* Granger test achieves significance. The pattern is consistent with the theoretical prediction: cross-border contagion propagates with temporal lags (through exposure networks and policy responses across jurisdictions), introducing exactly the linear temporal structure that Granger tests are designed to detect. The quantile test fails ($p = 0.492$), confirming that the predictive relationship is concentrated in the upper tail—consistent with a threshold exceedance rather than a location shift.

Region decomposition. The regional AUROC breakdown (Table 10) reveals that Asia (0.803) and Europe (0.685) drive the global signal, while the U.S. sub-panel alone achieves only 0.390. The low U.S. sub-panel AUROC reflects the small $N = 6$ and shortened time frame ($T = 76$ vs. $T = 140$ for the full U.S. bank-level analysis); the baseline U.S. bank-level pipeline with $N = 30$ over $T = 140$ achieves 0.805 (Table 8). The high Asian AUROC likely reflects the dramatic credit/GDP expansion in China and Japan that preceded and accompanied the GFC—structural shifts that are well-captured by the BSDT distance-from-normal framework.

8.11.2 Synthesis Across Scales

Table 11 compares the three empirical configurations.

Table 11: Cross-scale synthesis: Baseline MFLS performance at three levels of aggregation. All results use real data only.

Configuration	N	T	AUROC	GFC Lead	Granger
Institution-type (FDIC aggregated)	7	140	≥ 0.70	6Q	FAIL
Bank-level (FDIC individual)	30	140	0.805	6Q	FAIL
G-SIB cross-border (real data)	20	76	0.781	6Q	PASS

Three patterns emerge across scales: (i) the 6-quarter GFC lead is *invariant* to aggregation level, data source, and geography—a structural property of the BSDT operator, not a data artefact; (ii) AUROC improves from 0.70 (7 institution types) to 0.81 (30 banks), and the cross-border panel achieves 0.78 despite coarser country-level data—confirming that the signal is robust across heterogeneous data sources; (iii) Granger causality transitions from universal failure (domestic panels) to partial success (cross-border), consistent with the theoretical prediction that international contagion introduces temporal lags that create linear predictability absent in domestic phase transitions.

9 Welfare Calibration

9.1 CCyB Calibration Formula

Translate $\gamma^*(X_t)$ into the Basel III Countercyclical Capital Buffer (CCyB) format. Under a log-linear approximation around steady state, the required buffer increment is:

$$\Delta\text{CCyB}(t) \approx \kappa^{-1} \cdot \gamma^*(X_t) \cdot \sigma_\ell(t), \tag{9}$$

where $\sigma_\ell(t)$ is the cross-sectional standard deviation of leverage in the banking system at time t, observable quarterly from BIS data. Equation (9) converts the abstract coupling coefficient into basis points, directly comparable to the Basel III CCyB range of 0–250 bps.

9.2 Welfare Cost of Inaction

The expected welfare loss from running at $\gamma = 0$ over an instability window $[t_0, t_1]$ is:

$$\mathcal{L}_{\text{inaction}} = \int_{t_0}^{t_1} \left[E(X_t) - E(X_t^{\gamma^*})\right] dt.$$

Converting to consumption-equivalent units via a standard Euler equation ($C_t \propto e^{-E(X_t)/(\sigma\theta)}$), the percentage loss in permanent consumption is:

$$\%CL = 100 \times \left(1 - e^{-\mathcal{L}_{\text{inaction}}/(\sigma\theta T)}\right).$$

Table 12 reports the calibrated welfare losses for the three crisis windows on real FDIC data. The GFC 2008 window ($T = 10$ quarters of elevated energy above the median) produces a consumption-equivalent loss of 19.57 percentage points, consistent with the consensus range of macro studies [14]. COVID 2020 shows near-zero welfare loss in the model because the energy spike dissipated within two quarters under emergency policy response; the CCyB path nevertheless peaks at 231 bps, reflecting the severity of the initial shock. The 2022 rate-shock episode produces a modest loss of 1.22 pp, consistent with a stress event absorbed without systemic transmission.

Table 12: Welfare calibration results on real FDIC data ($T = 140$, $N = 7$, $d = 6$). $\mathcal{L}$: consumption-equivalent welfare loss (percentage points). Peak CCyB: maximum implied buffer increment (basis points). See Section 9 for calibration details.

Crisis Window	$\mathcal{L}$ (pp)	Peak CCyB (bps)
GFC 2008 (2007-Q3–2009-Q4)	19.57	152
COVID 2020 (2019-Q4–2021-Q2)	0.00	231
Rate Shock 2022 (2022-Q1–2023-Q4)	1.22	0

Notes. COVID $\mathcal{L} = 0$ because no quarter in that window exceeded the reference-period median plus two consecutive quarters of above-threshold energy; the unprecedented policy response compressed the crisis duration below the model's persistence threshold. The COVID CCyB peak of 231 bps reflects that the *fragility level* was extreme even though the welfare loss was averted. The 2022 CCyB = 0 bps reflects that $\gamma^*(t)$ was near zero in that window: the rate shock was absorbed by institution-level adaptation without a systemic energy spike.

9.3 Pigouvian Interpretation

From Section 2, the optimal tax is $\tau_i^* = \gamma^*(X)\,\partial E/\partial \ell_i$. The macro-prudential instrument is therefore a *gradient-indexed levy*: the surcharge on institution i is proportional to how much its marginal leverage change increases the system energy. Institutions with high connectivity (large $\partial E/\partial \ell_i$) pay more; institutions in isolation pay less. The total tax $\gamma^*(X)\,\|\nabla E(X)\|^2$ equals the welfare gap $V_{\bar{\gamma}} - V^*$ to leading order, confirming the Pigouvian interpretation: the tax exactly corrects the externality.

10 Discussion

10.1 Summary of Empirical Contributions

The empirical analysis, conducted at three levels of aggregation on exclusively real data, produces eight findings:

1. **Super-criticality is the norm.** The U.S. banking system spends 68.6% of the 1990–2024 sample above the critical manifold $\mathcal{C}$. Crises do not push the system above $\mathcal{C}$; they exploit a system already in a super-critical state.

2. **Anti-alignment reveals complementarity.** The gradient alignment between E_{BS} and Φ is $\cos\theta = -0.335$ on real data. This negative alignment indicates that detection (∇E_{BS}) and restoration ($-\nabla\Phi$) measure *geometrically perpendicular* axes of instability, making their combination more informative than either alone.

3. **Scale-invariant 6-quarter GFC lead.** The MFLS alarm fires 6 quarters before Lehman across all three configurations: institution-type ($N = 7$), bank-level ($N = 30$), and cross-border G-SIB ($N = 20$)—a structural property of the BSDT operator, not a data artefact.

4. **Universal domestic Granger failure.** Five independent MFLS variants—spanning unsupervised, polynomial, logistic, and gated families—all fail Granger causality at $\alpha = 0.10$ on domestic U.S. data. The failure is structural: crises are phase transitions, not trends, and linear temporal predictability is the wrong test.

5. **Cross-border Granger success.** On the $N = 20$ G-SIB panel, threshold Granger causality *passes* ($p = 0.043$), and linear Granger is borderline significant ($p = 0.053$). International contagion propagates with temporal lags that introduce exactly the linear structure Granger tests detect.

6. **Precision-recall frontier.** On the $N = 7$ panel, variants trade off monotonically between hit rate and selectivity, tracing a Pareto curve that separates "distance from equilibrium" from "probability of transition."

7. **Herding discovery.** The Signed LR variant learns a negative weight ($\beta_{\delta_T} = -1.38$) on temporal novelty: before crises, institutions become *more* predictable from their own history. This is the quantitative signature of convergent herding.

8. **Cross-border robustness.** The Baseline AUROC on the $N = 20$ G-SIB panel (0.781) is comparable to the domestic bank-level result (0.805), despite using coarser country-level data for non-U.S. institutions—confirming that the BSDT signal generalises across data sources and geographies.

10.2 Limitations

First-order dynamics. GravityEngine implements first-order gradient flow; it has no velocity state or mass matrix. The proofs in this paper are calibrated to this model. An extension to second-order (inertial) dynamics would require replacing the Lyapunov candidate $V = \Phi(X)$ with $V = \Phi(X) + \frac{1}{2}\dot{X}^\top M\dot{X}$ and re-deriving the stability rate.

Homogeneous feature space. All institutions are modelled in the same d-dimensional space. Heterogeneous institution types (commercial banks vs. insurance companies vs. hedge funds) have different natural feature spaces. The framework extends by using a block-diagonal covariance Σ_0 and separate deviation operators per class, but the equivalence constants c_1, c_2 will depend on the block structure.

Static network. The critical manifold $\mathcal{C}$ depends on $\lambda_{\max}(\mathbf{W})$, the spectral radius of the network. The results assume a fixed $\mathbf{W}$; in practice, the network topology changes over time (especially during crises). Extending Theorem 5.3 to a time-varying $\mathbf{W}(t)$ is an open problem.

Non-U.S. data granularity. The cross-border G-SIB extension (Section 8.11) uses country-level banking-sector aggregates from the World Bank GFDD for non-U.S. institutions, not individual bank balance sheets. This is a coarser approximation than the FDIC individual-bank data available for U.S. G-SIBs. Institution-level international data (e.g., Bankscope, S&P Capital IQ) would improve the cross-border analysis but require commercial data licences.

Short G-SIB calibration window. The G-SIB panel begins in 2005 (limited by World Bank GFDD coverage), yielding only 8 quarters of normal-period calibration data (2005–2006). This short window inflates the false alarm rate to 88.7% under a P75 threshold. LOCO robustness checks show that FAR drops to 19–27% when the threshold is calibrated excluding individual crisis windows, indicating that the high FAR is primarily a threshold-calibration artefact rather than a signal-quality issue.

Granger causality. The domestic Granger failure, while theoretically consistent with the phase-transition thesis, means that the model cannot be validated by standard linear forecasting metrics on U.S.-only data. The partial success on the cross-border panel ($p = 0.043$, threshold Granger) suggests that international contagion introduces temporal structure absent in domestic phase transitions, but replication on larger cross-border panels is needed to confirm this.

10.3 Open Problems

1. **Global Lyapunov extension.** Theorems 5.2 and 5.3 are local (linearisation-based). A global Lyapunov argument for the full nonlinear system above $\mathcal{C}$ would require controlling

higher-order terms in the Gronwall bound—the force-clamp (Appendix A8) provides the regularisation needed, but the argument is not yet complete.

2. **Heterogeneous equilibria.** The uniqueness of $X^\star$ is not proved; the erf-log pairwise potential is non-convex and admits multiple local minima. Understanding the basin structure is important for policy: which equilibrium the system converges to after a crisis depends on initial conditions.

3. **Data-driven separation constant.** Assumption E4 (separation condition $\nu > 0$) is verified empirically but not derived from data statistics. A data-driven lower bound on ν from the spectral properties of the BSDT operator Jacobian would make the theorem fully quantitative.

4. **Nonlinear forecasting tests.** The threshold Granger success on the G-SIB panel ($p = 0.043$) is a first step. Exceedance regression [10], quantile causality, and time-varying parameter models should be systematically applied across all three panel configurations to map the boundary between phase-transition (Granger-immune) and contagion (Granger-detectable) regimes.

5. **Real-time estimation.** The current analysis uses full-sample Ledoit–Wolf shrinkage. Deploying the framework in real time requires recursive covariance estimation and rolling BSDT re-calibration with a specified look-back window; the computational cost is $O(N^2 d)$ per quarter, which is operationally feasible.

6. **Institution-level international data.** Replacing World Bank country aggregates with institution-level balance-sheet data (Bankscope, S&P Capital IQ, or ECB IBSI microdata) would improve the cross-border analysis and enable institution-level CCyB recommendations for non-U.S. G-SIBs.

10.4 Policy Implications

For central banks to adopt the CCyB formula (9) in practice, three conditions must be met: (1) the network adjacency matrix $\mathbf{W}$ must be observable with quarterly frequency (currently available from BIS bilateral exposures for 30 countries); (2) the deviation operators δ_k must be re-calibrated annually on updated normal-period data (computationally tractable at $O(N^2)$ per re-calibration); and (3) the intervention cost κ must be estimated from GDP data (the ratio of growth lost per unit of credit restriction). All three are operationally feasible with existing BIS and FRED data infrastructure.

The herding discovery (finding 6) has a direct policy implication: a macro-prudential indicator should weight *decreasing* behavioural diversity as a warning sign, not just increasing distance from equilibrium. The current Basel III credit-to-GDP gap measures only aggregate leverage deviation; the MFLS framework adds cross-sectional *convergence* as a crisis predictor.

Appendices

A Full Proofs of Theorems 1–6 and C

We provide complete proofs for all theorems in the main text. Proofs are ordered by increasing generality.

Proof of Theorem 5.1 (Monotone Descent, full). Define $V(t) = \Phi(X(t))$. Differentiating: $\dot{V} = \nabla\Phi(X)^\top \dot{X} = \nabla\Phi^\top(-\nabla\Phi) = -\|\nabla\Phi\|^2 \leq 0$. Since $V(t)$ is non-increasing and bounded below by $\Phi(X^\star)$, it converges to $V_\infty \geq \Phi(X^\star)$. By LaSalle, every ω-limit point X_∞ satisfies $\nabla\Phi(X_\infty) = 0$. *Discrete case*: backtracking line search in `gravity.py` (lines 659–670) explicitly rejects steps that increase energy; the force clamp at $F_{\max} = 100$ prevents numerical overflow. □

Proof of Theorem 5.2 (Local Exponential Stability, full). Write $z(t) = X(t) - X^\star$ with $\|z\|$ small. Expanding: $\dot{z} = -\nabla\Phi(X^\star + z) = -H^\star z + O(\|z\|^2)$. Since $H^\star \succ 0$ (Assumption E2), all eigenvalues of $A^\star = -H^\star$ have real part $-\lambda_i(H^\star) < 0$. By Lyapunov's indirect method, $X^\star$ is locally exponentially stable with rate $r = \lambda_{\min}(H^\star)$.

Gronwall bound: decompose $\dot{z} = -\alpha z + \Delta_{\text{pair}}$ with $\|\Delta_{\text{pair}}\| \leq L_{\text{pair}} \|z\|$. Then: $\frac{d}{dt}\|z\|^2 \leq -2\alpha\|z\|^2 + 2L_{\text{pair}}\|z\|^2 = -2(\alpha - L_{\text{pair}})\|z\|^2$. For $\kappa = \alpha - L_{\text{pair}} > 0$, $\|z\|^2(t) \leq \|z\|^2(0)e^{-2\kappa t}$. □

Proof of Theorem 5.3(ii) (Growing Mode, full). Above $\mathcal{C}$, let u be the unit eigenvector of $D^2\Phi_{\text{pair}}(X^+)$ corresponding to $\lambda_u > \alpha$. Define $W(t) = \frac{1}{2}\langle X(t) - X^\star, u\rangle^2$. Differentiating: $\dot{W} = \left\langle \dot{X} - 0, u \right\rangle \langle X - X^\star, u\rangle = \left\langle -\alpha(X - X^\star) - D^2\Phi_{\text{pair}}(X - X^\star) + O(\|z\|^2), u \right\rangle \langle X - X^\star, u\rangle$. Projecting onto u: $\dot{W} = (-\alpha + \lambda_u)W + O(\|z\|^3) = cW + O(\|z\|^3)$ with $c = \lambda_u - \alpha > 0$. By forward Gronwall: $W(t) \geq W(0)e^{ct/2}$ for small t. This grows exponentially; no fixed $\bar{\gamma}$ prevents it (increasing $\bar{\gamma}$ shifts $\mathcal{C}$ but there always exist configurations above the shifted $\mathcal{C}$ for small enough pairwise distances). □

Proof of Proposition 6.1 (OGD Regret, full). Let $\Delta_\gamma = \gamma_{\max} - \gamma_{\min}$. Standard OGD [3] with convex c_t and step $\eta_\gamma = \Delta_\gamma/\sqrt{T}$: $R_T \leq \frac{\Delta_\gamma^2}{2\eta_\gamma} + \frac{\eta_\gamma}{2}\sum_t \|\nabla_\gamma c_t\|^2 \leq \frac{\Delta_\gamma\sqrt{T}}{2} + \frac{L_c^2\Delta_\gamma T}{2\sqrt{T}} = L_c\Delta_\gamma\sqrt{T}$.

Oracle approximation by power iteration: maintain unit vector v_t; update $\tilde{v}_{t+1} = (\nabla\Phi_{\text{pair}}(X^t + hv_t) - \nabla\Phi_{\text{pair}}(X^t))/h$, renormalize. After K steps, the Rayleigh quotient $\hat{\lambda}_t$ satisfies $|\hat{\lambda}_t - \lambda_{\max}| \leq \lambda_{\max}(\lambda_2/\lambda_{\max})^{2K}$. Choosing $K = O(\log N)$ gives approximation error $O(N^{-1})$. □

Proof of Theorem 7.2 (Optimal Policy, full derivation). First-order condition of the Bellman equation w.r.t. γ: $0 = 2\kappa\gamma + \beta\,\partial_\gamma\mathbb{E}[V(X') \mid X, \gamma]$. Using $\partial_\gamma X' = -\eta\nabla E(X)$ and $\partial_{X'}V \approx \nabla E(X')/(1-\beta)$ (first-order approximation): $\partial_\gamma\mathbb{E}[V] \approx -\eta/(1-\beta)\langle\nabla E(X'), \nabla E(X)\rangle$. At leading order $\nabla E(X') \approx \nabla E(X)$, giving $2\kappa\gamma^* \approx \beta\eta\|\nabla E\|^2/(1-\beta)$, so $\gamma^* \approx \frac{\beta\eta\|\nabla E\|^2/(1-\beta)}{2\kappa}$. The saturation factor $E/(E+\theta)$ with $\theta = \kappa/\beta$ ensures $\gamma^* \to 0$ as $E \to 0$ (no intervention needed in the normal regime). □

Proof of Lemma (Force-Clamp Regularisation – Appendix to Proof 7). The clamped force $F_{\text{clamp}} = \Pi_{B(F_{\max})}(F_{\text{raw}})$ is the projection onto the closed ball of radius $F_{\max}$. Projection onto a convex set is non-expansive: $\|F_{\text{clamp}}(X) - F_{\text{clamp}}(Y)\| \leq \|F_{\text{raw}}(X) - F_{\text{raw}}(Y)\|$. On any sublevel set $\mathcal{L}_c$, log-repulsion coercivity gives minimum pairwise distance $\delta_{\min}(c) > 0$, so the Jacobian

norm of F_{raw} is bounded by $F_{\max}/\delta_{\min}(c) < \infty$. The contraction condition $\alpha > F_{\max}/\delta_{\min}(c)$ is achievable for any finite c by increasing α. □

B Computational Details

B.1 GravityEngine Parameter Table

Parameter	Symbol	Default value	Role
Radial spring	α	0.1	Mean-reversion strength
Pairwise coupling	γ	1.0	Interaction scale
Gaussian range	σ	1.0	Attraction length-scale
Repulsion coefficient	λ	0.1	Prevents collapse
Softening	ε	10^{-5}	Regularises log-repulsion
Force clamp	$F_{\max}$	100	Bounds pairwise Jacobian
Step size	η	0.02	Euler step
Backtracking factor	—	0.5	Armijo line-search halving

B.2 Pairwise Force Kernel

The erf-based pairwise potential between agents i and j at distance $r = \|x_i - x_j\|$ is:

$$\phi(r) = \gamma \frac{\sigma\sqrt{\pi}}{2} \operatorname{erf}\left(\frac{r}{\sigma}\right) - \gamma\lambda \log(r + \varepsilon).$$

The attraction force $-\phi'(r)\hat{r}$ is Gaussian: $\phi_1'(r) = \gamma e^{-r^2/\sigma^2}$. The repulsion force is long-range logarithmic: $\phi_2'(r) = \gamma\lambda/(r + \varepsilon)$. The net force is attractive at long range and repulsive at short range, giving a stable cluster configuration at the equilibrium distance $r^* = \sigma\sqrt{\log(1/\lambda)}$ (implicit solution of $\phi'(r^*) = 0$).

B.3 Finite-Difference HVP

The Hessian-vector product $D^2\Phi_{\text{pair}}(X)v$ required by the power iteration is computed via the symmetric finite difference:

$$D^2\Phi_{\text{pair}}(X)v \approx \frac{\nabla\Phi_{\text{pair}}(X + hv) - \nabla\Phi_{\text{pair}}(X - hv)}{2h}, \quad h = 10^{-4}.$$

This requires two gradient evaluations ($O(N^2 d)$ each) per power iteration; the approximation error is $O(h^2 \left\|D^3\Phi\right\|)$, negligible for $h = 10^{-4}$.

C Data Sources and Processing

C.1 FDIC Statistics on Depository Institutions (SDI)

All banking features (f_0–f_4) are drawn from the FDIC SDI quarterly call reports, accessed via the public REST API at `https://banks.data.fdic.gov/api/financials`. Data are aggregated by `SPECGRP` (institution type: 7 categories) and queried at individual quarter-end dates (e.g., `filters=REPDTE:20070331`). The six call-report fields used are: `ASSET` (total

assets), `LNLSNET` (net loans and leases), `EQ` (total equity capital), `NCLNLS` (noncurrent loans and leases), `NETINC` (net income), `EINTEXP` (total interest expense—note: this is the correct FDIC field name, not `INTEXP`).

C.2 Federal Reserve FRED Series

Feature f_5 (yield-curve slope) is the 10-Year minus 2-Year Treasury Constant Maturity spread (series T10Y2Y), downloaded from FRED at quarterly frequency.

Benchmark stress indices used for lead-lag analysis:

- STLFSI: St. Louis Fed Financial Stress Index (weekly, averaged to quarterly)
- VIX: CBOE Volatility Index (daily close, averaged to quarterly)
- NFCI: Chicago Fed National Financial Conditions Index (weekly, averaged to quarterly)
- T10YFF: 10-Year Treasury minus Fed Funds Rate (spread used in SRISK-proxy composite)

C.3 Normal-Period Definition

The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters). This span is chosen to be: (a) post-S&L resolution (completed 1993); (b) pre-housing boom (credit growth accelerated 2004); (c) long enough for stable covariance estimation ($T_{\text{ref}} = 40 \gg d = 6$). All BSDT operator parameters (μ_0, Σ_0, PCA loadings, velocity thresholds, KDE bandwidths) are fitted on this reference period only.

C.4 Replication

The full replication package, including all data-loading scripts, pipeline code, figure-generation scripts, and cached API responses, is available at `https://github.com/segetii/fintech` (branch `sync/merge-2025-10-22`). The pipelines can be reproduced with:

```
# Institution-type pipeline (N=7, T=140)
cd research/adaptive-friction/upgraded/pipeline
python run_pipeline.py

# Bank-level pipeline (N=30, T=140)
cd research/adaptive-friction/banklevel
python run_pipeline_banklevel.py

# G-SIB cross-border pipeline (N=20, T=76, real data only)
cd research/adaptive-friction/banklevel_enhanced
python run_pipeline_gsib_real.py

# All 5 MFLS variants on G-SIB panel
python run_variants_gsib_real.py --n_boot 500
```

D Additional Crisis Window: 2022 Rate Shock

The 2022 rate-shock window (2022-Q1 through 2023-Q4) is characterised by the Federal Reserve's 525-basis-point policy rate increase, the fastest tightening cycle since 1980. The MFLS signal shows a moderate energy spike beginning 2022-Q2, but crucially, the OOS alarm (trained on P75 of pre-2007 data) does *not* fire during this window: the energy level remains below the GFC-calibrated threshold.

This is consistent with the model: the 2022 stress was publicly anticipated via forward guidance, allowing institutions to adapt. The welfare loss is modest ($\mathcal{L} = 1.22$ pp) and the implied CCyB is zero. The model correctly distinguishes between a predictable repricing event (2022) and an abrupt structural crisis (2008).

E G-SIB Cross-Border Data Sources

The G-SIB panel (Section 8.11) uses three data sources:

E.1 FDIC SDI (U.S. G-SIBs)

The 6 U.S. G-SIBs are queried at institution level (by FDIC CERT number) from the FDIC SDI REST API. Features: loan-to-asset ratio (`LNLSNET/ASSET`), equity ratio (`EQ/ASSET`), NPL ratio (`NCLNLS/LNLSNET`), ROA (`NETINC/ASSET`), and funding cost (`EINTEXP/LNLSNET`). Quarterly frequency, 2005-Q1 through 2023-Q4.

E.2 World Bank Global Financial Development Database (GFDD)

Non-U.S. G-SIBs use country-level banking-sector aggregates from the World Bank GFDD, accessed via the World Bank API v2 (`api.worldbank.org/v2/country/{CC}/indicator/{ID}`). Indicators used:

Feature	World Bank Indicator	Coverage
Equity ratio	FB.BNK.CAPA.ZS	2005–2023 (most countries)
NPL ratio	FB.AST.NPER.ZS	2005–2023 (most countries)
ROA	GFDD.SI.01	2000–2021
Credit-to-GDP	GFDD.DI.01	2000–2021
Lending rate	FR.INR.LEND	2000–2023 (non-euro-area)

All World Bank indicators are annual. Interpolation to quarterly frequency uses cubic spline (`scipy.interpolate.CubicSpline`) with not-a-knot boundary conditions. Countries covered: GBR, FRA, DEU, ITA, NLD, JPN, CHN.

E.3 ECB Monetary and Financial Institution Interest Rates (MIR)

For euro-area G-SIBs (BNP Paribas, Société Générale, Deutsche Bank, UniCredit, ING), bank lending rates to non-financial corporations are obtained from the ECB Statistical Data Warehouse (`data-api.ecb.europa.eu/service/data/MIR/`). Series: `M.{CC}.B.A2A.A.R.A.2240.EUR.N` (monthly, converted to quarterly by end-of-quarter selection). Countries: DE, FR, IT, NL. The ECB MIR does not cover UK institutions (HSBC, Barclays, Standard Chartered); for these, the World Bank `FR.INR.LEND` lending rate is used instead.

E.4 Data Limitations

Non-U.S. entries are *country-level banking-sector aggregates*, not individual institution balance sheets. Two U.S. G-SIBs (Morgan Stanley, State Street) are excluded because their FDIC CERT numbers return zero coverage in the 2005–2023 date range (they hold FDIC certificates for small subsidiary banks, not the holding company). Japan and China have partial coverage gaps in some indicators (2005–2009 for CHN equity ratio; 2022–2023 for JPN ROA); gaps are forward/backward-filled before interpolation.

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